

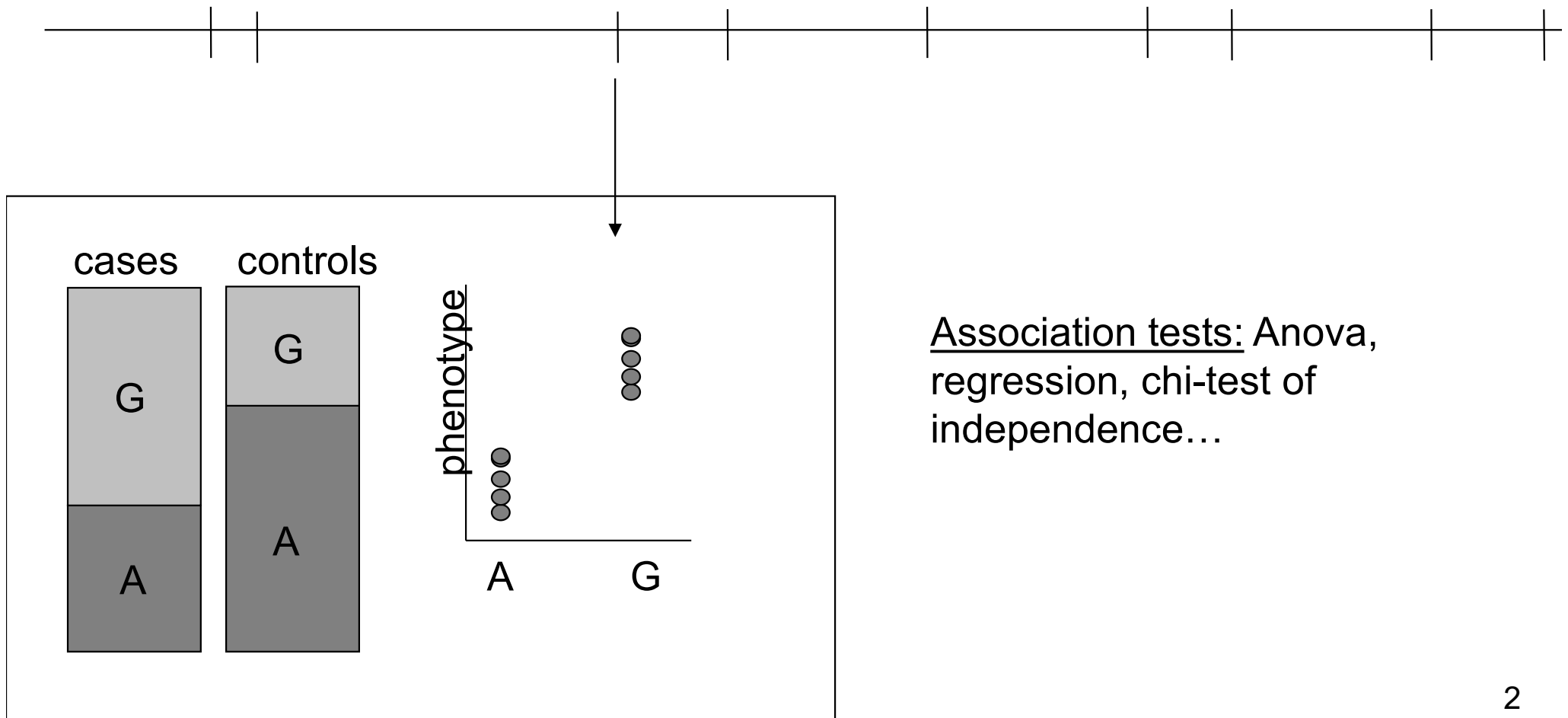
Systems Genetics

Spring 2025

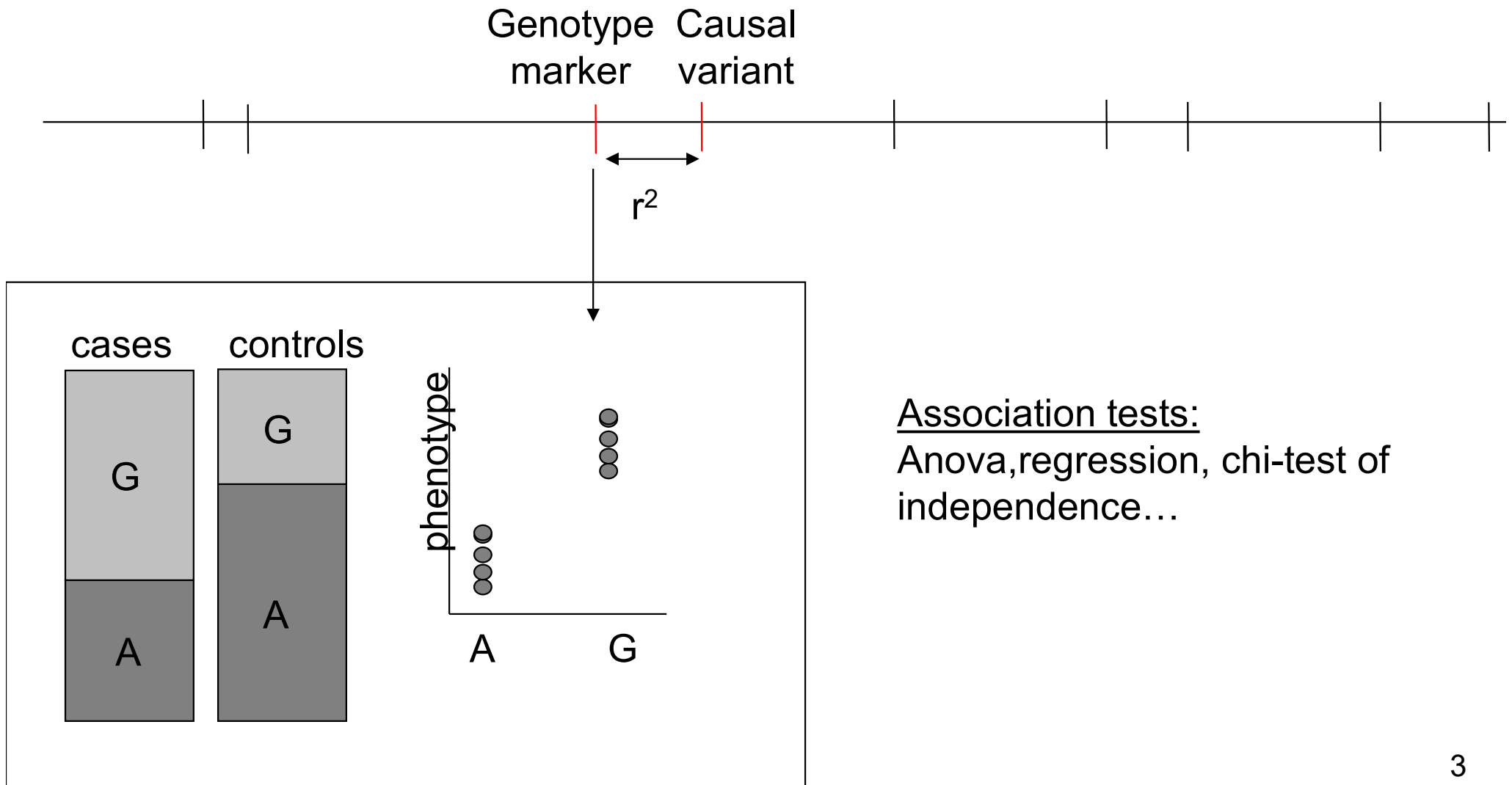
Lecture 7

GWAS – part 2

Testing each marker across the genome



Indirect association (based on linkage disequilibrium)



Statistical power

$$\mathbb{E}[\chi^2] \propto N\gamma^2 p(1 - p)r^2,$$

Fixed by nature:

effect size γ^2

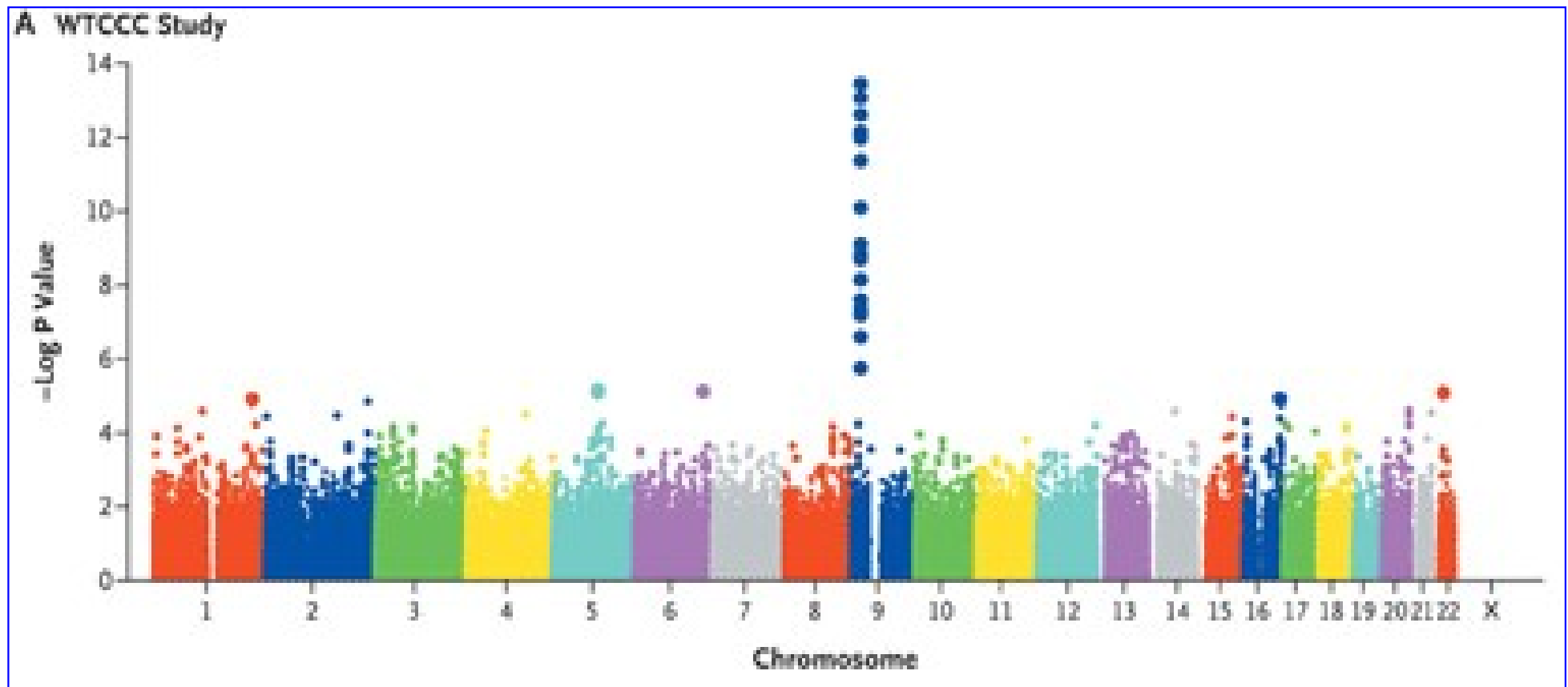
Allele frequency p

Non-Fixed parameters:

Correlation r^2

Sample size N

-Log₁₀ P Values for SNP Associations with Myocardial Infarction

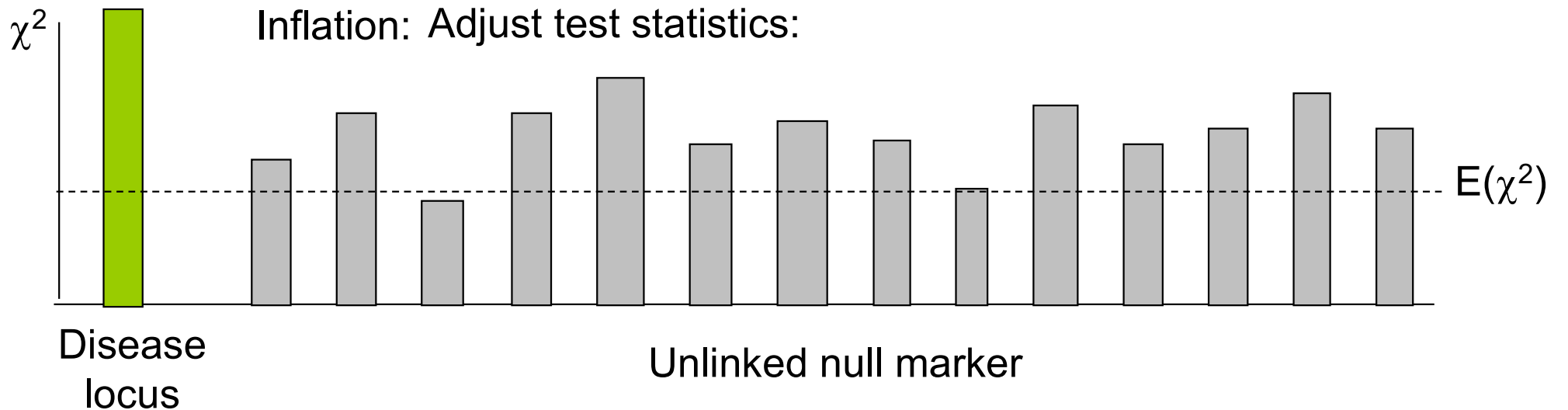
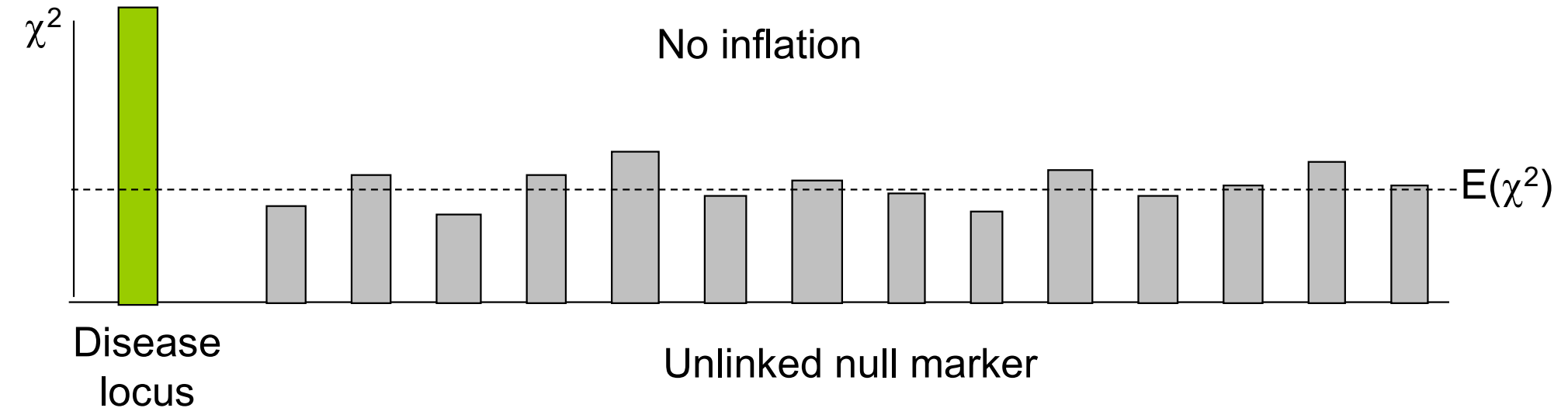


GWAS – the gory details

- How surprised should I be?
- Population stratification
- Representation of results

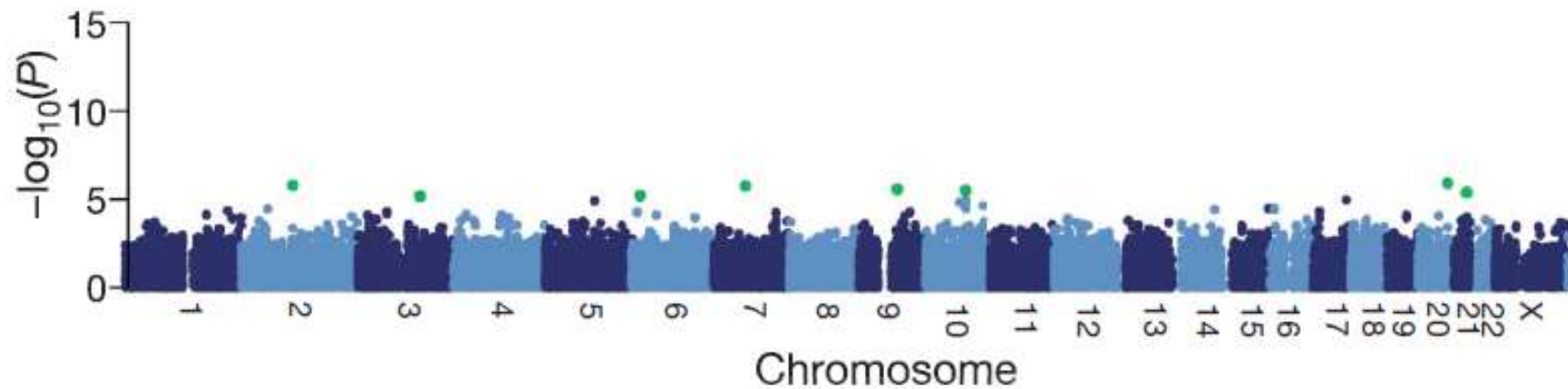
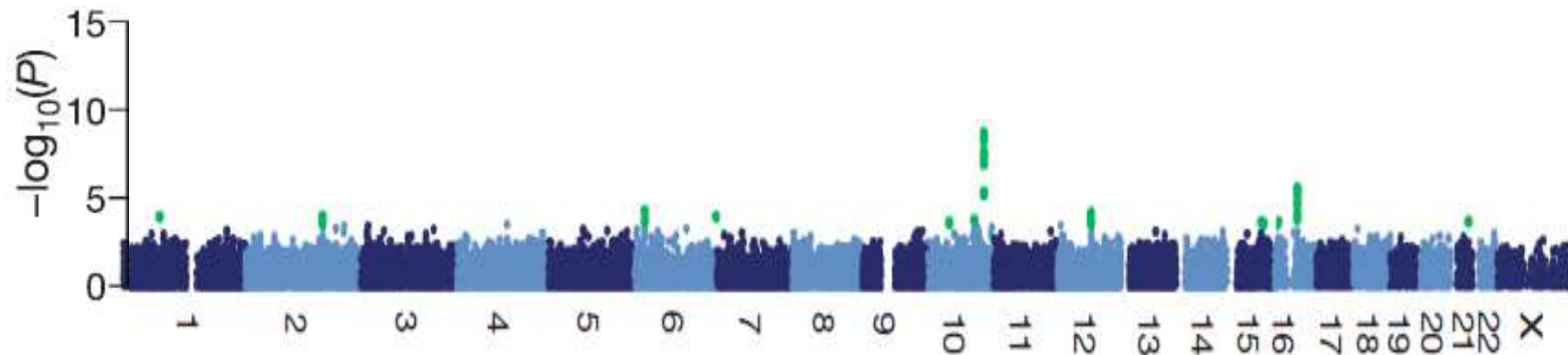
How surprised should I be?

Quantifying case-control biases

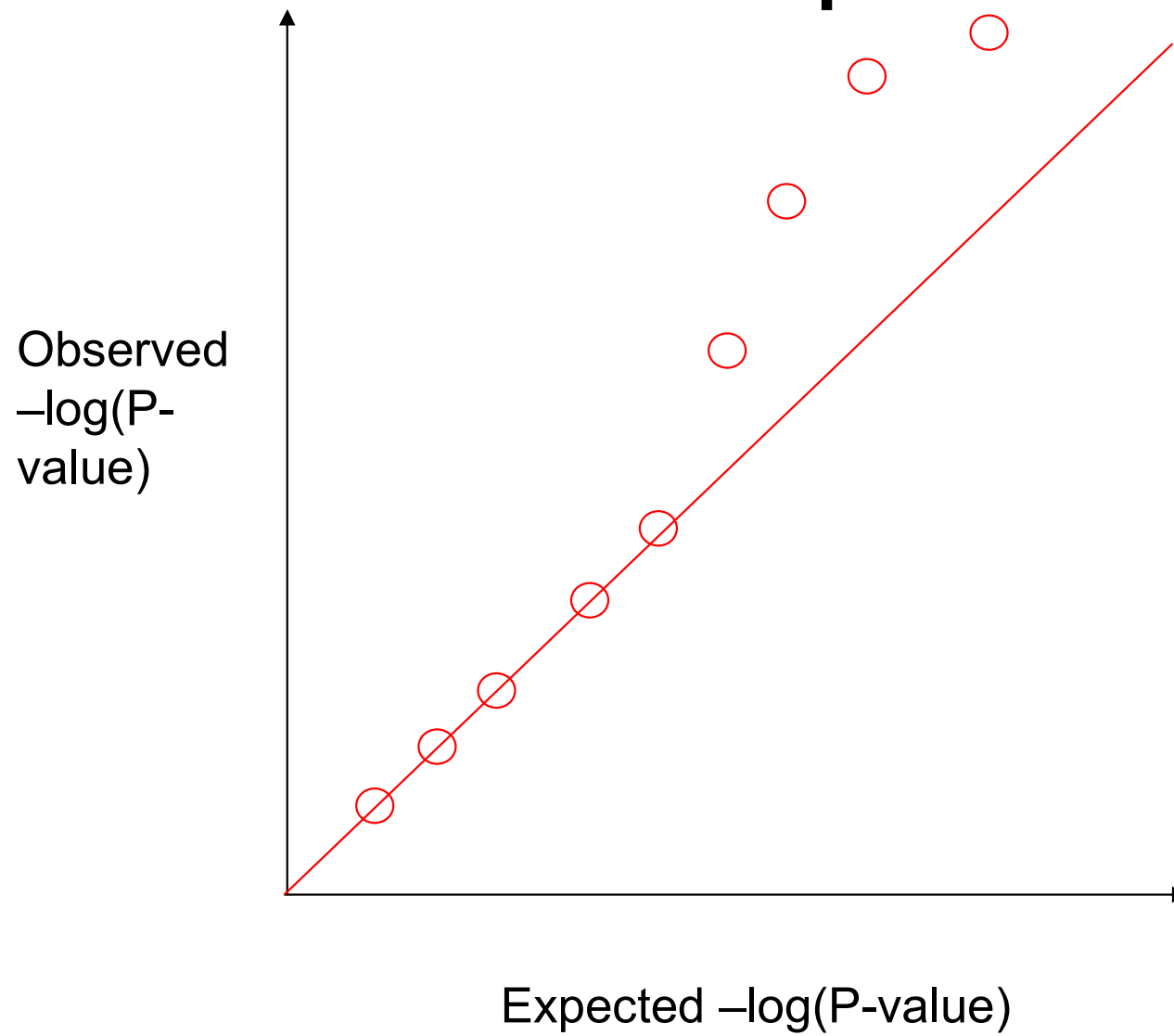


Genomic control $\lambda = \text{obs. median}(\chi^2) / \text{expected median}(\chi^2) = \text{Median}(\chi^2) / 0.455$

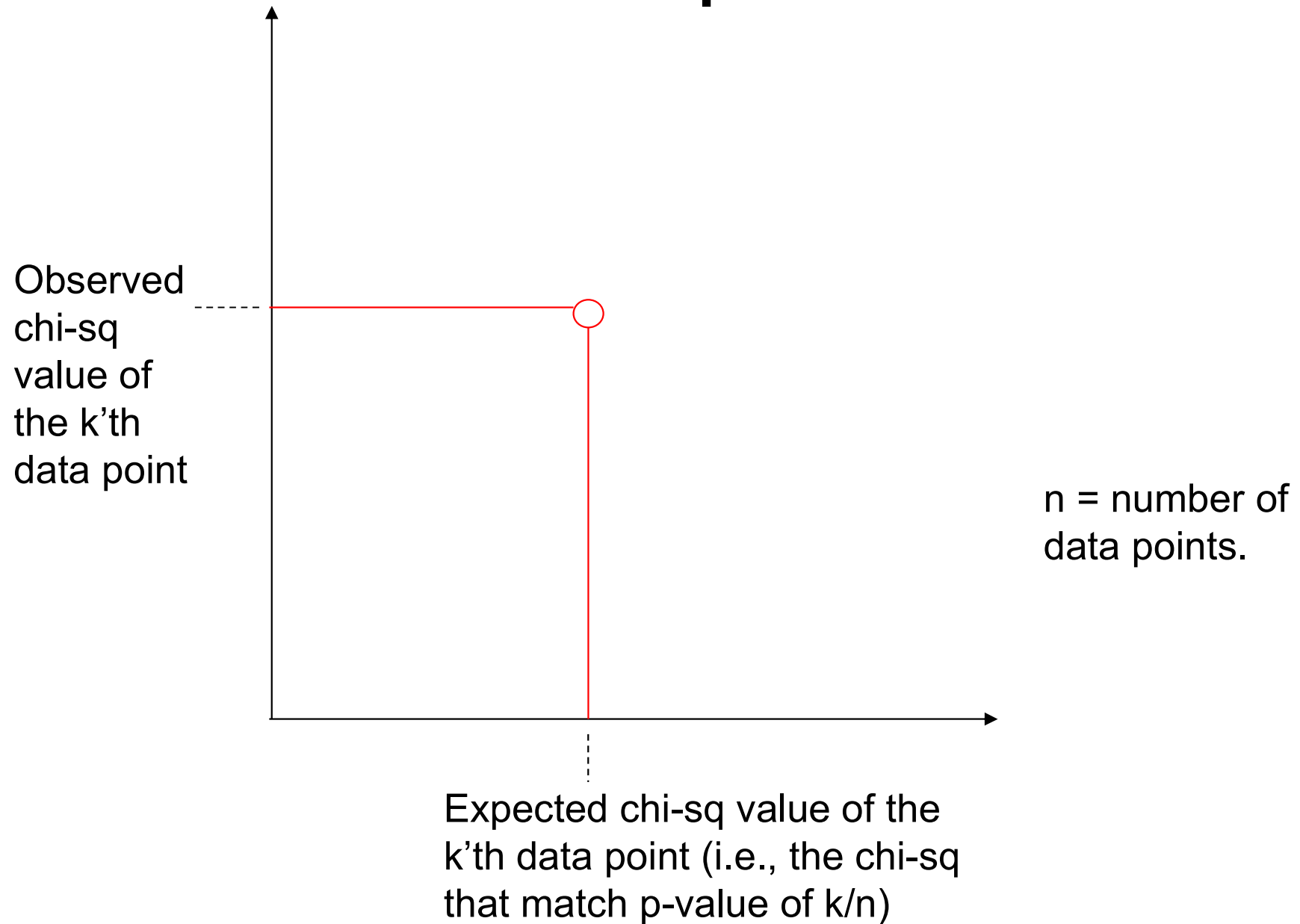
How surprised should I be?



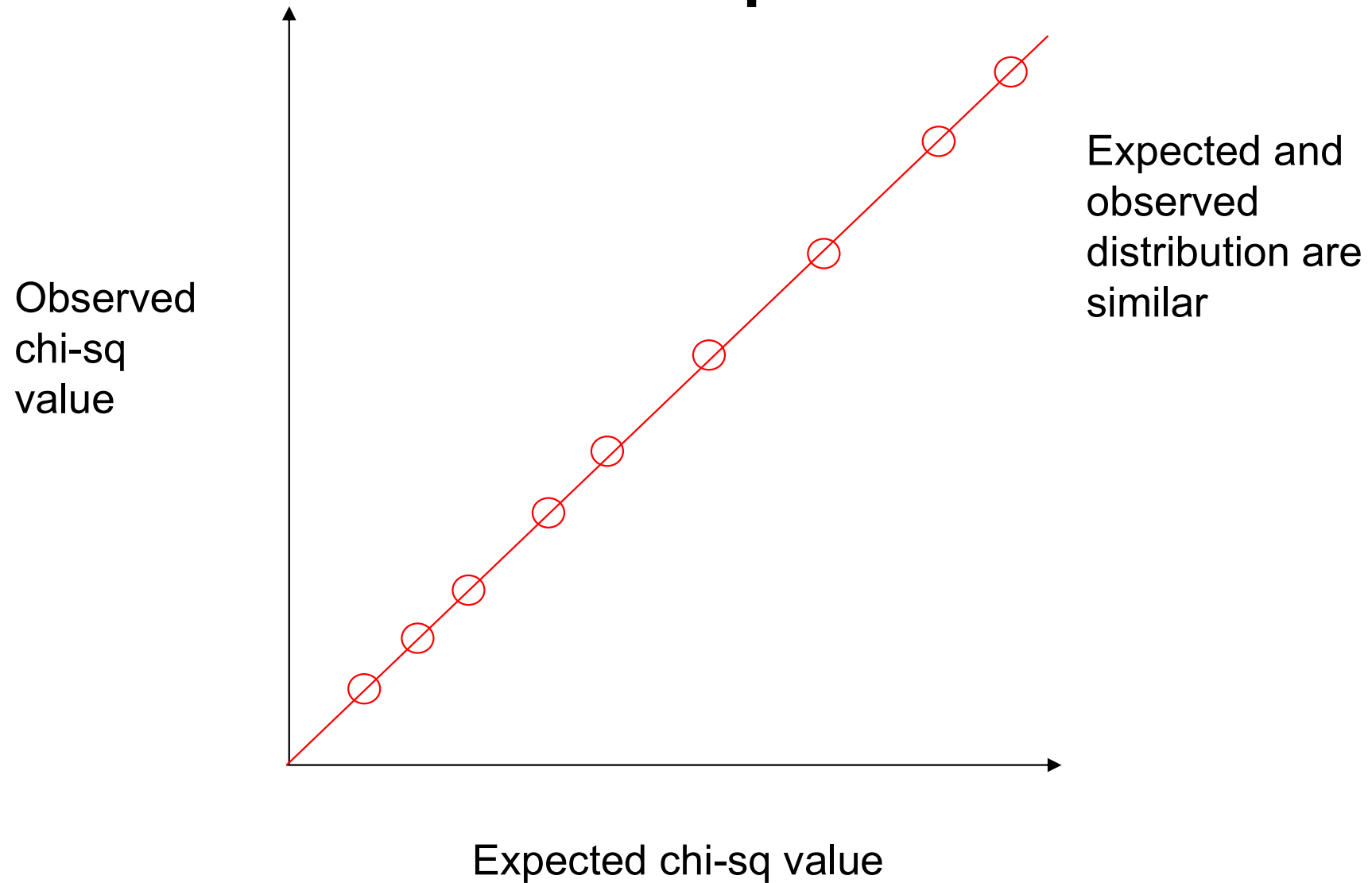
Q-Q plot



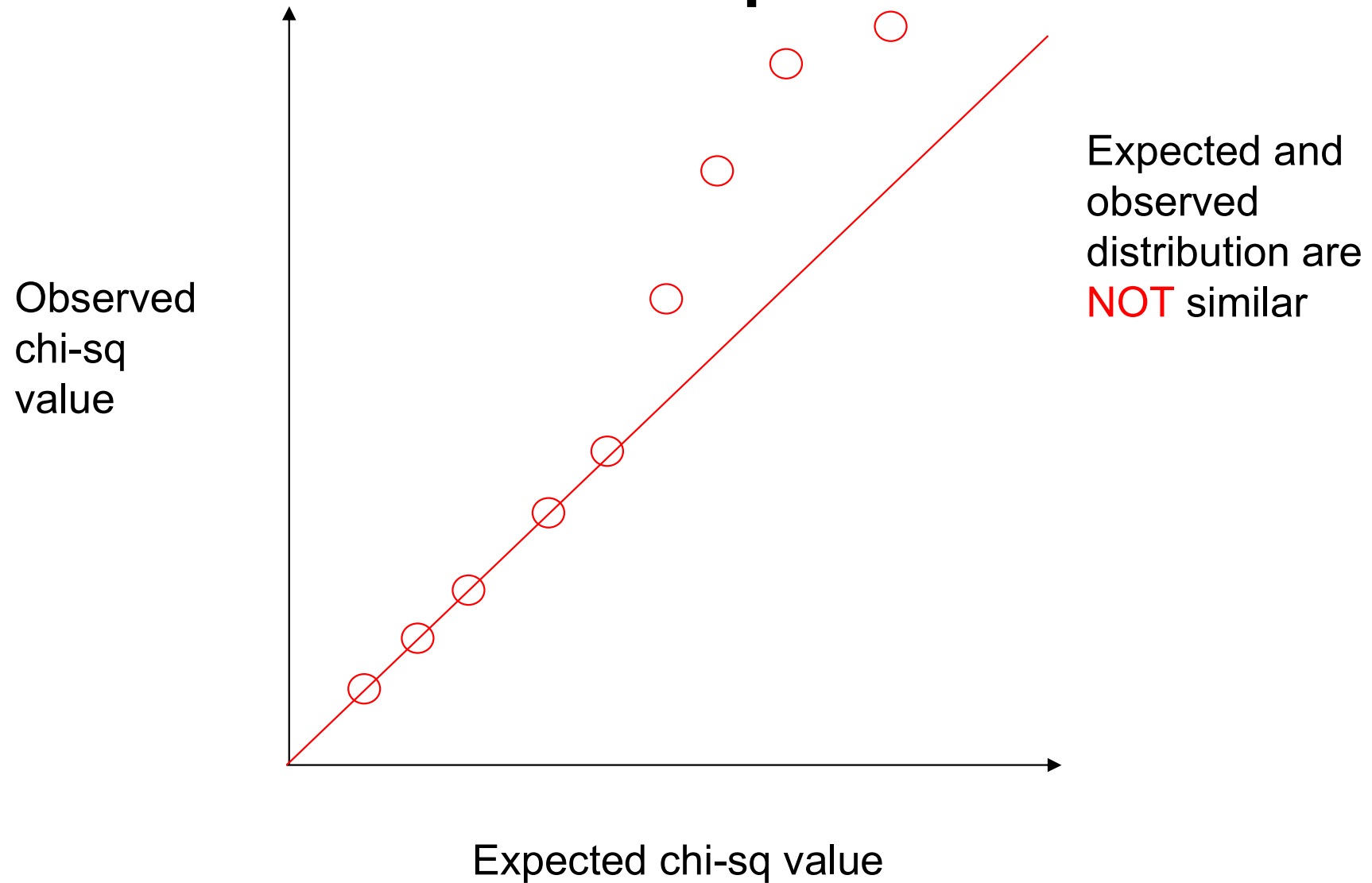
Q-Q plot



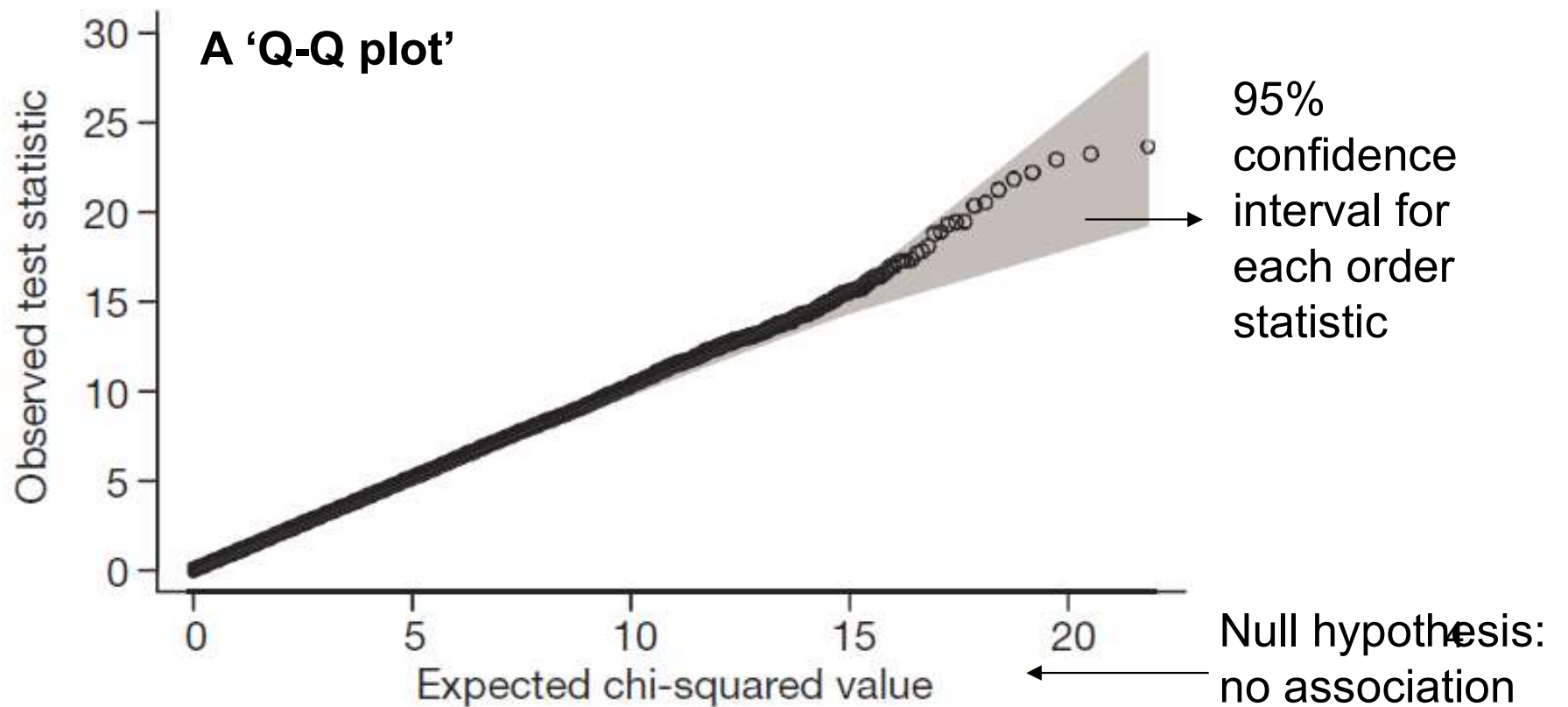
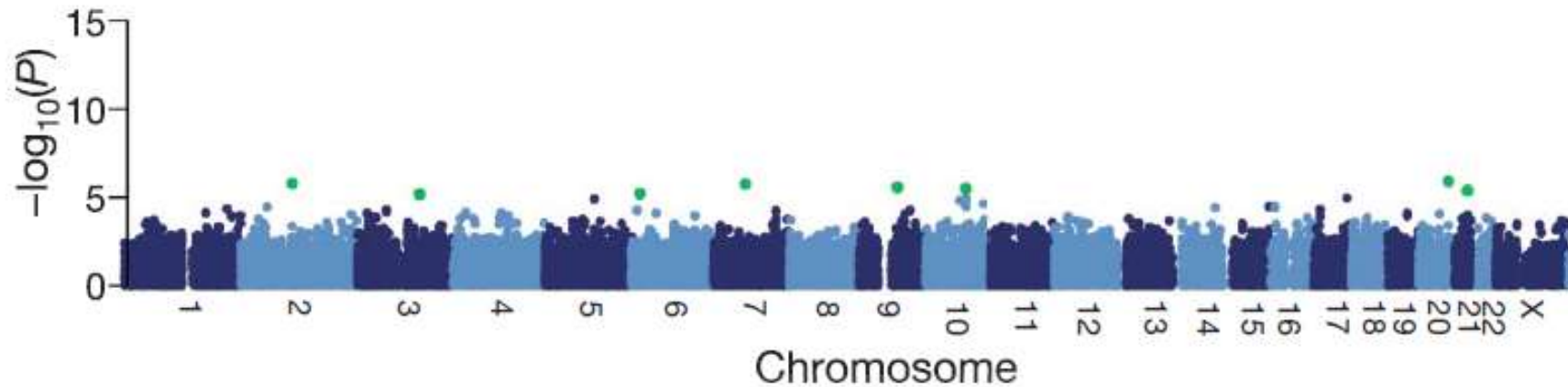
Q-Q plot



Q-Q plot

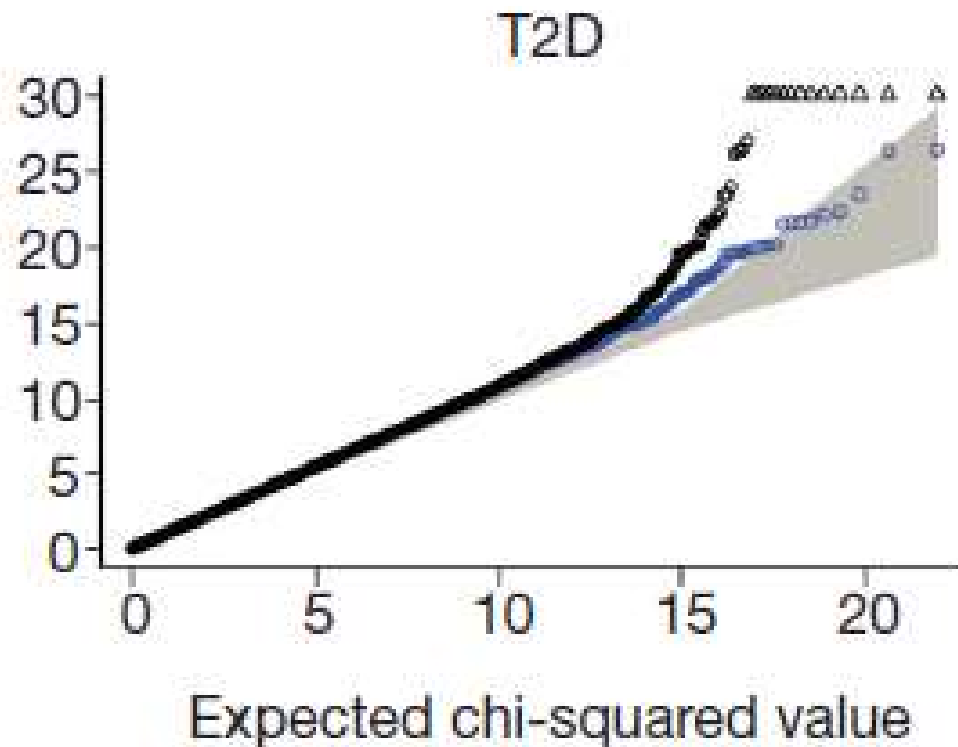
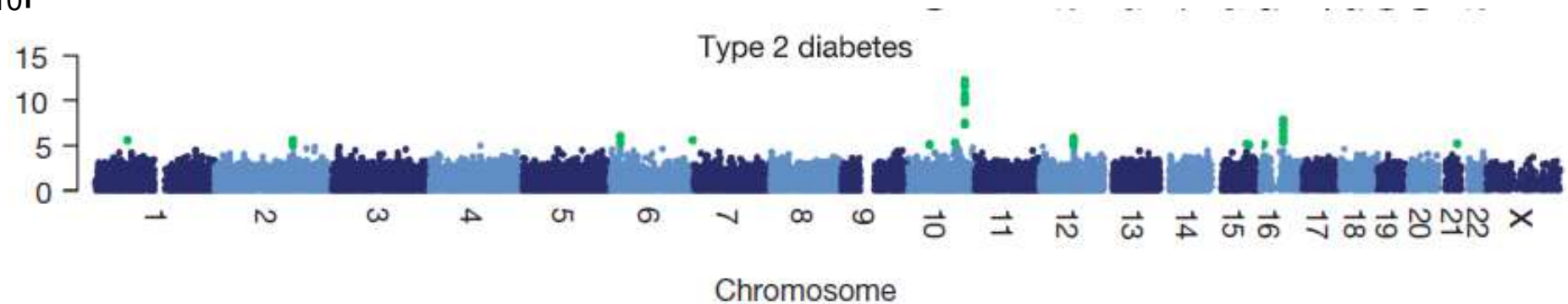


How surprised should I be?



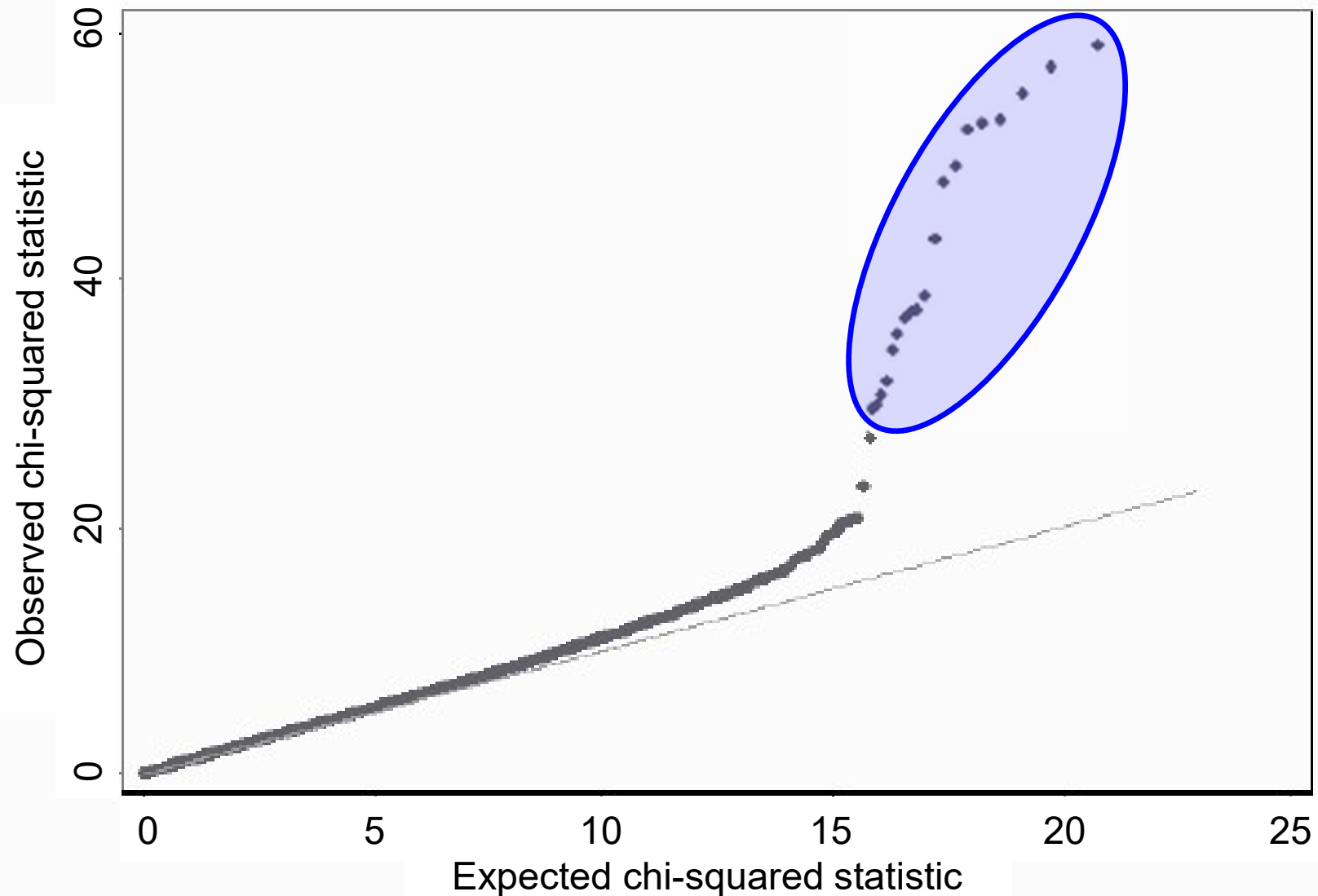
How surprised should I be?

$-\log_{10}p\text{-value}$

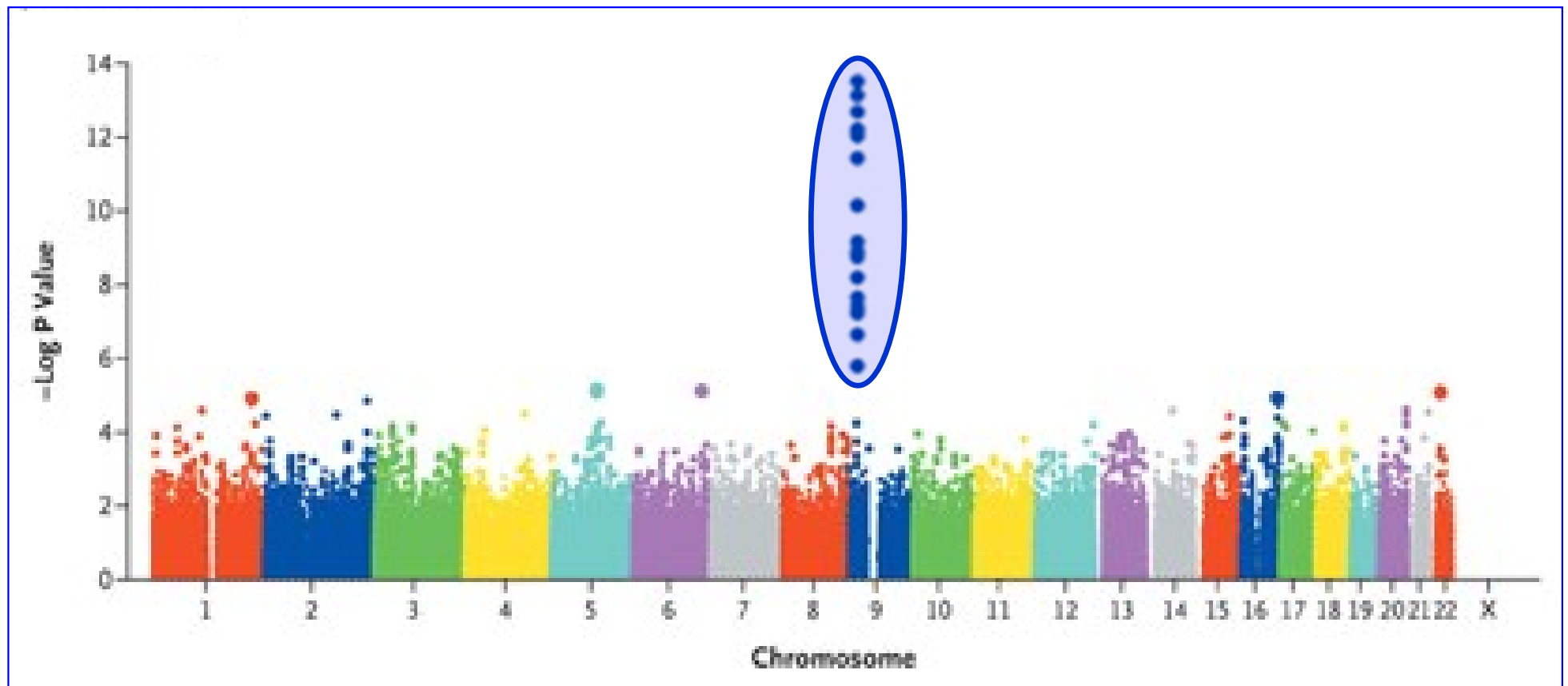


Myocardial Infarction

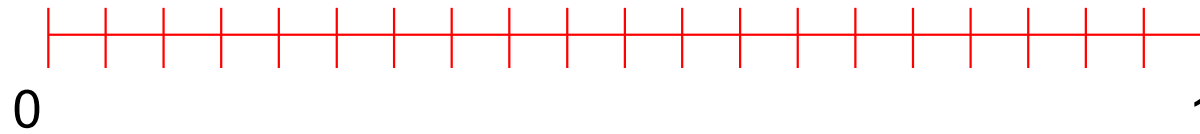
אוטם שריר הלב



$-\text{Log}_{10}$ P Values for SNP Associations with Myocardial Infarction



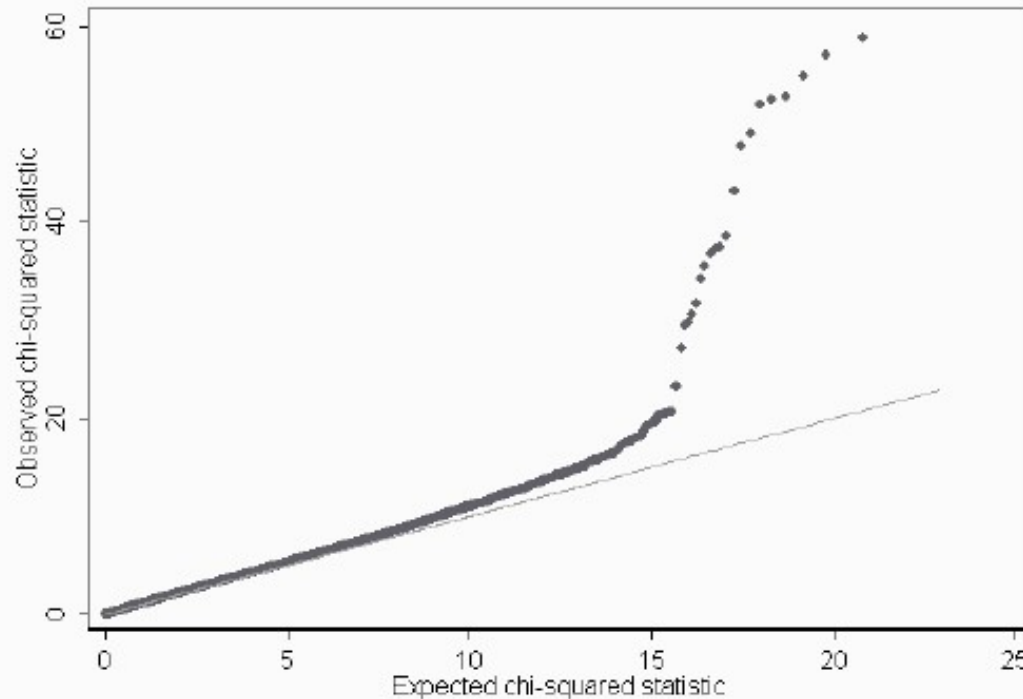
Under the null: p-values are uniformly distributed



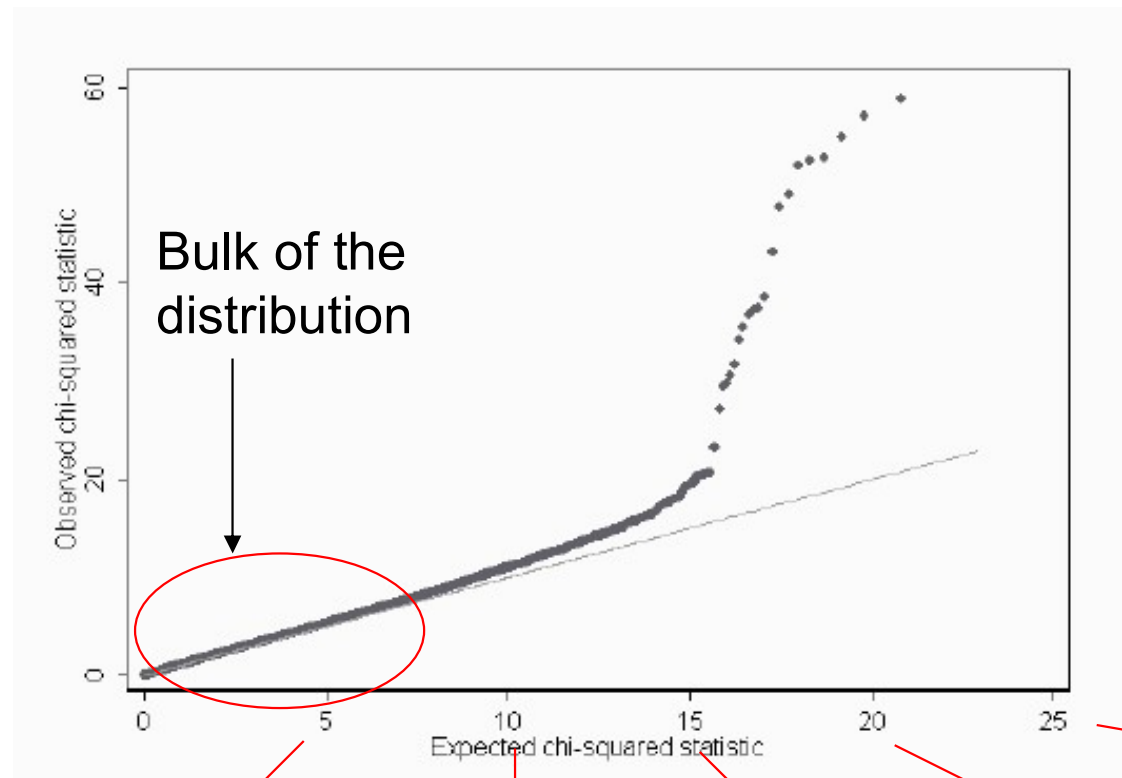
expected P-values



Expected χ^2 values



Under the null: p-values are uniformly distributed



pval=0.02

pval= 10^{-3}

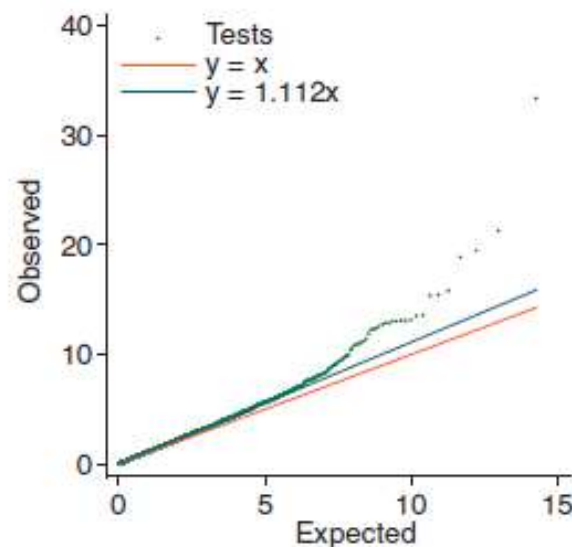
pval= 10^{-4}

pval= 10^{-5}

pval= 10^{-6}

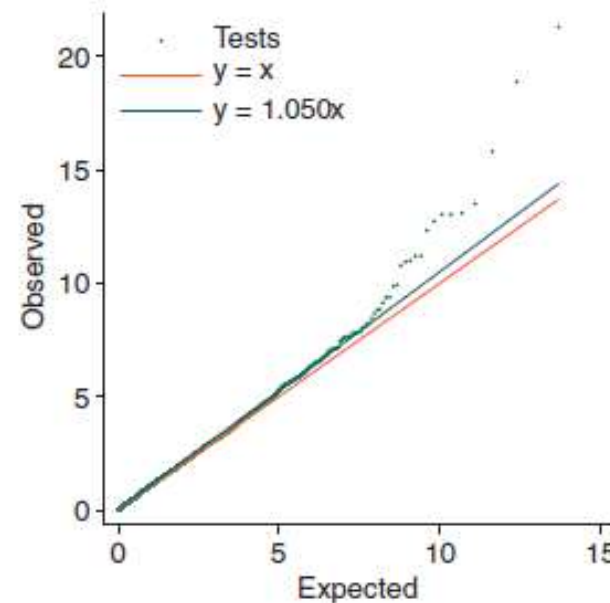
Genomic control - example

- Genomic control $\lambda = \frac{\text{median observed } \chi^2}{\text{median expected } \chi^2}$
- Gives a measure of whether the entire distribution is inflated $\lambda > 1$ or deflated $\lambda < 1$
- In most GWAS, 1.05 is good.



All 6322 SNPs

$\lambda = 1.112$

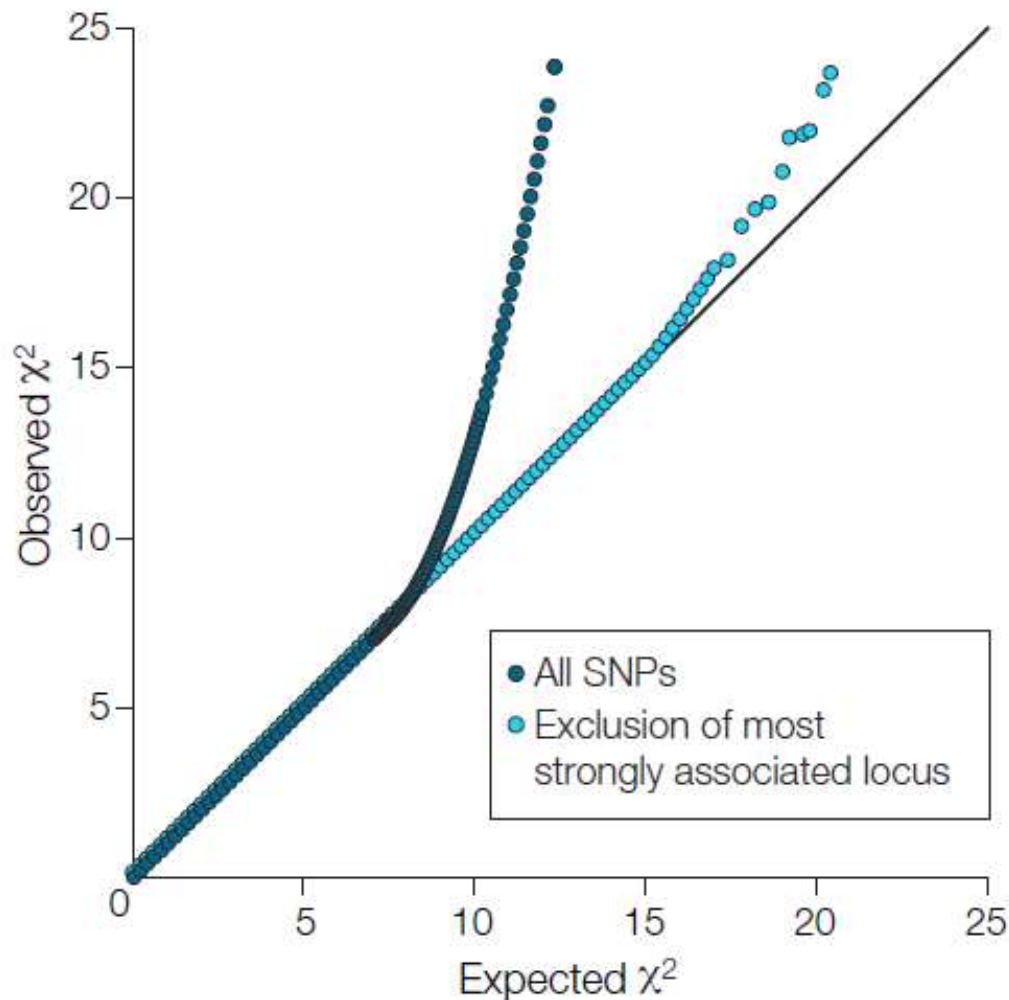


4629 SNPs with low difference in call rate between case and control.

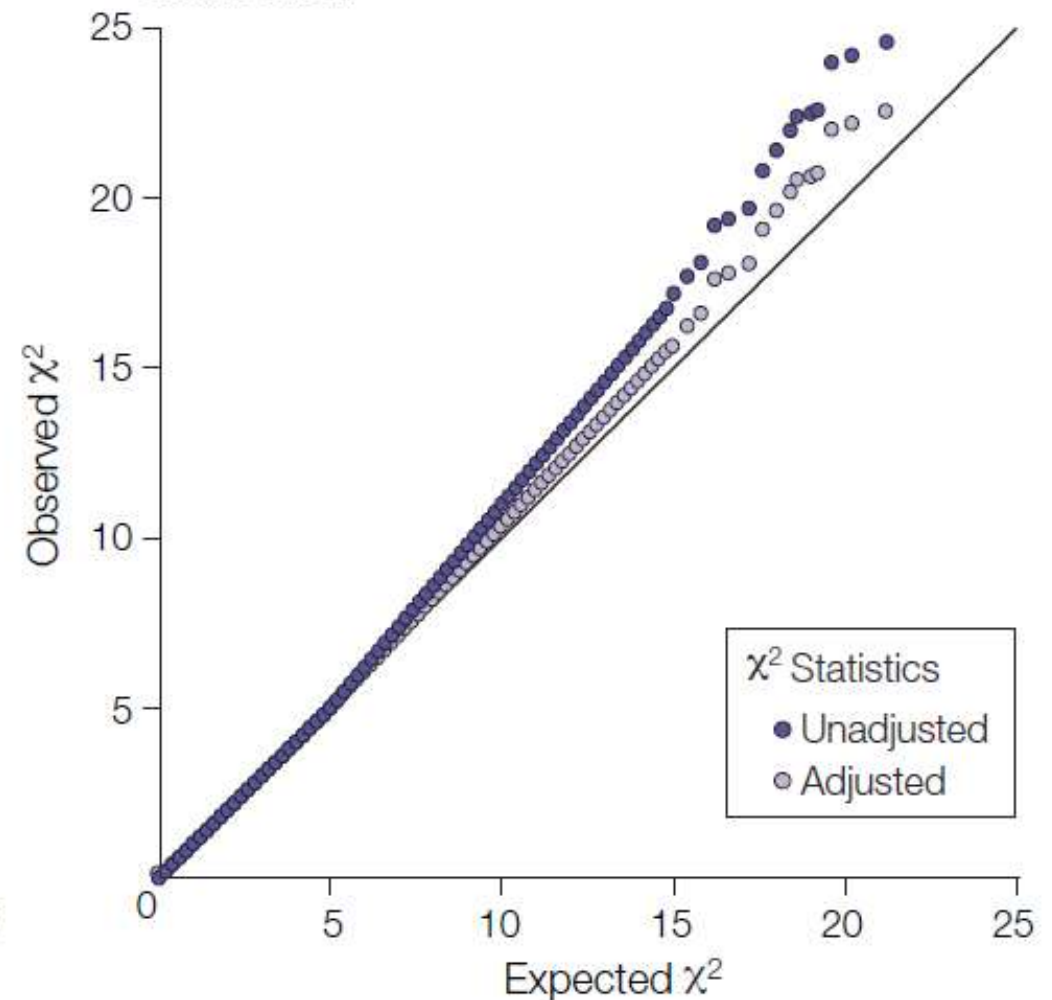
$\lambda = 1.05$

Q-Q plots: assess the number of associations

A Before-and-after exclusion of most strongly associated locus



B Before-and-after adjustment for population stratification



Representation of results

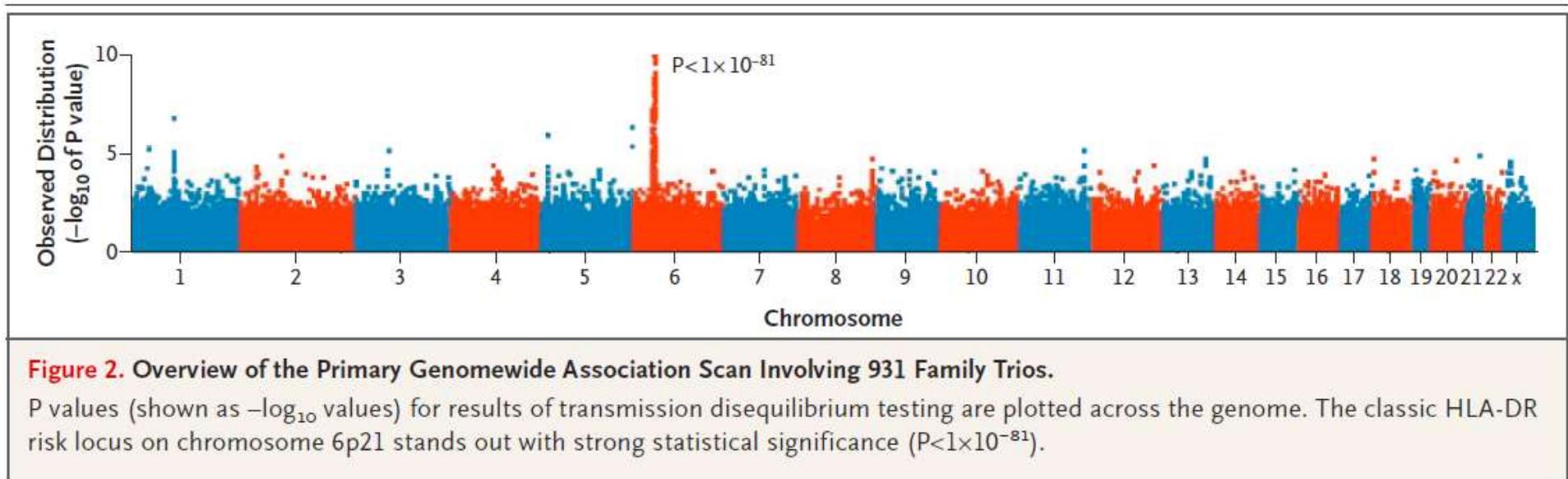
GWAS – the gory details

- How surprised should I be?
- Representation of results
- Population stratification

Risk Alleles for Multiple Sclerosis

(טרשת נפוצה)

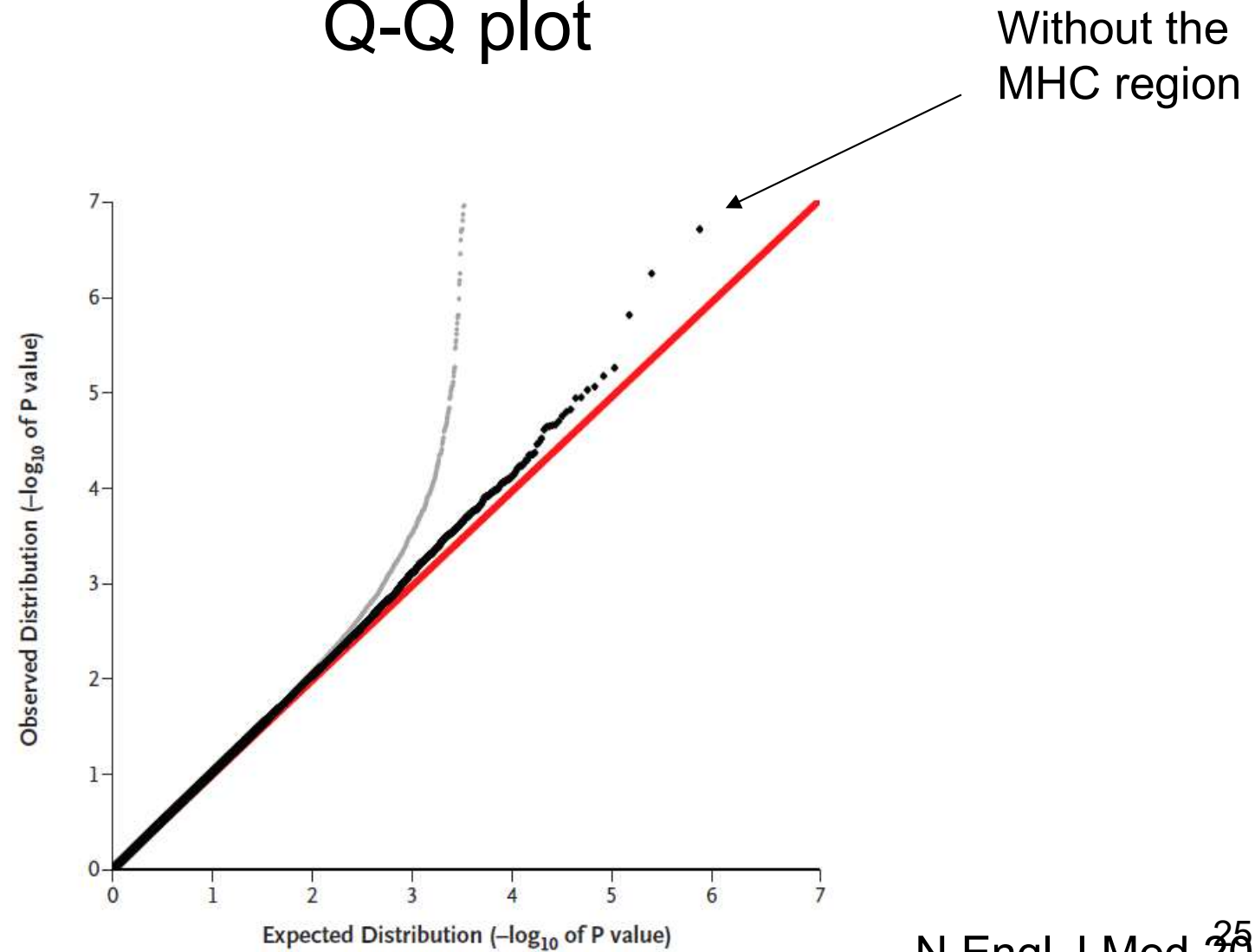
Manhattan plot



MHC region

Risk Alleles for Multiple Sclerosis

Q-Q plot

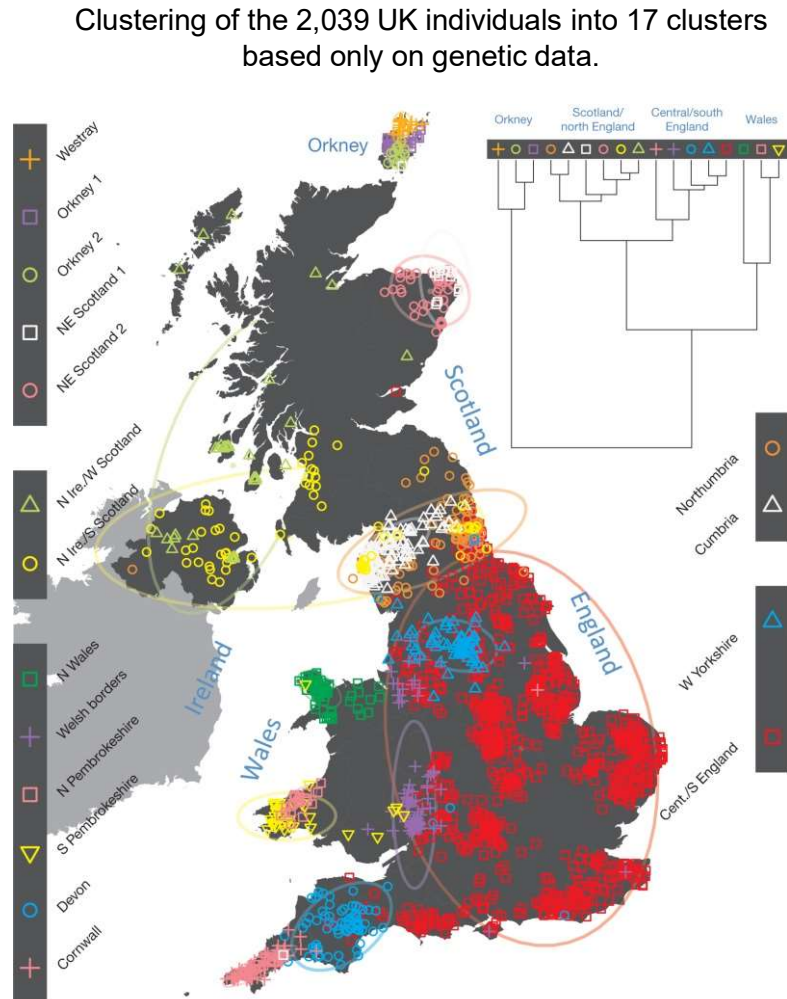


GWAS – the gory details

- How surprised should I be?
- Representation of results
- Population stratification

Population stratification

A rich pattern of subtle fine-scale genetic differentiation within the UK, which shows a marked concordance with geography (Leslie et al. Nature 2015)



Fine-scale genetic variation between human populations is interesting as a signature of historical demographic events and because of its **potential for confounding disease studies**

outline

1. What is population stratification
2. how to **measure** population stratification
3. How to **correct** for population stratification?

1. What is population stratification ?

מוצא ייחוס



Population stratification refers to **ancestry** differences between cases & controls in an association study.

Risk: spurious associations to markers with high ancestry differences.

Example: CF in the US population:

CF is prevalent in America-European population only - > differences in ancestry between case and controls. Due to different alleles between populations, there are spurious associations with no biological meaning.

Population stratification (cont)

Solution: restricting to a single ancestry.

- Example: association of height in European Americans.

The height trait is associated with the lactase SNP

phenotype

Tall



Short



Lactase SNP

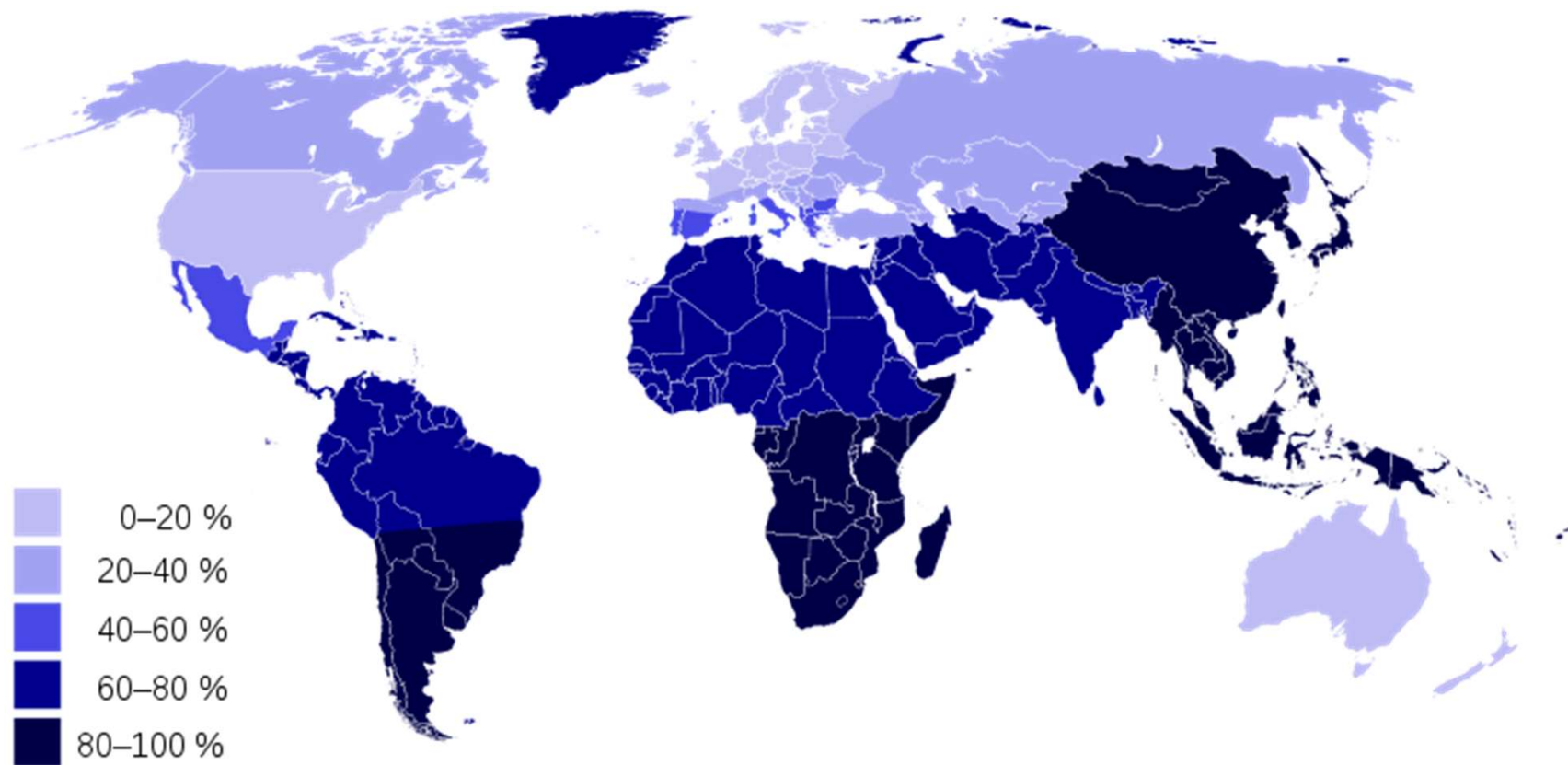
T

C

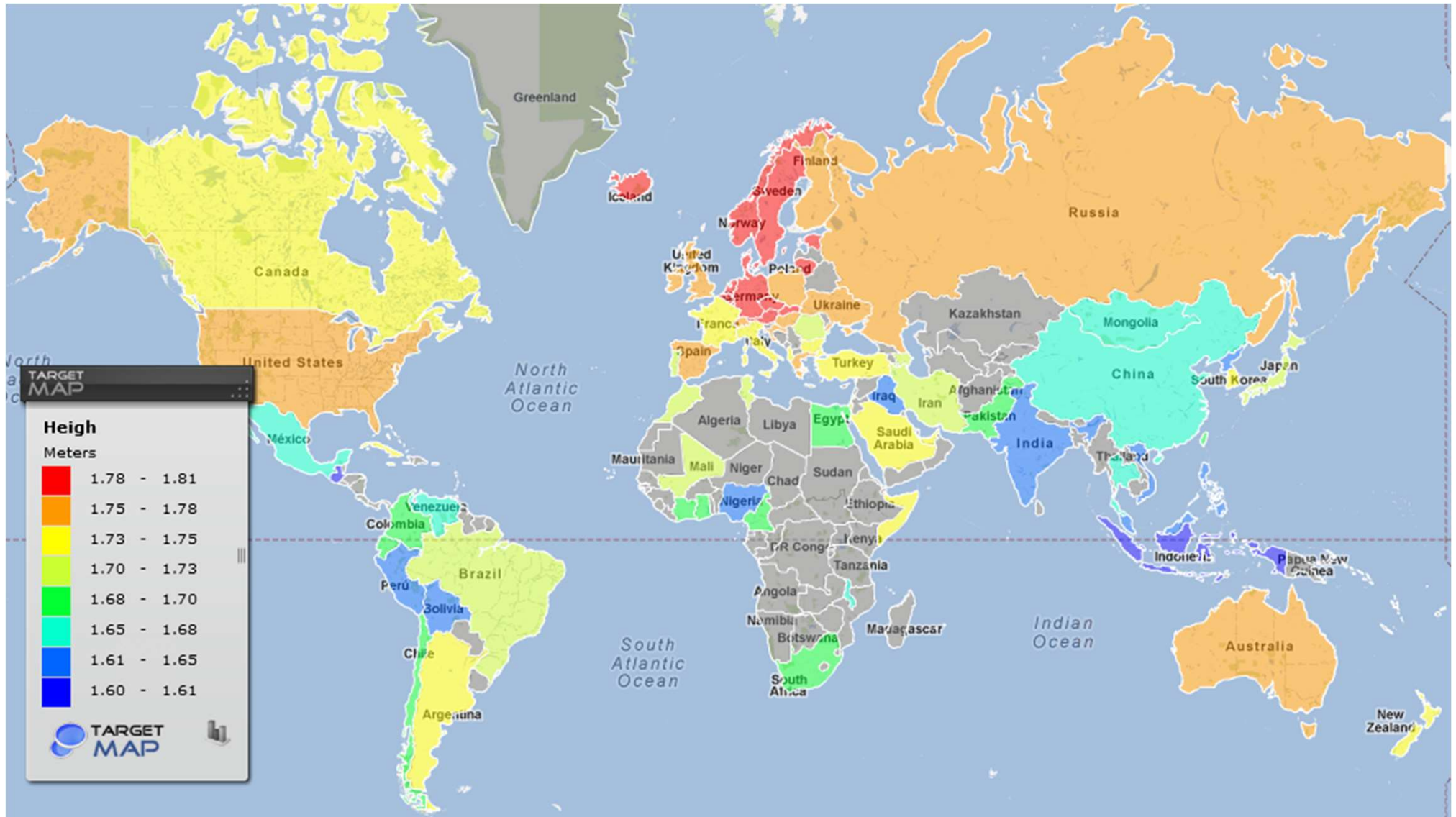
chr2

↓
The same SNP is
associated with lactase
activity in adults

The frequency of decreased lactase activity around the world



Average Male height map



Population stratification (cont)

Solution: restricting to a single ancestry.

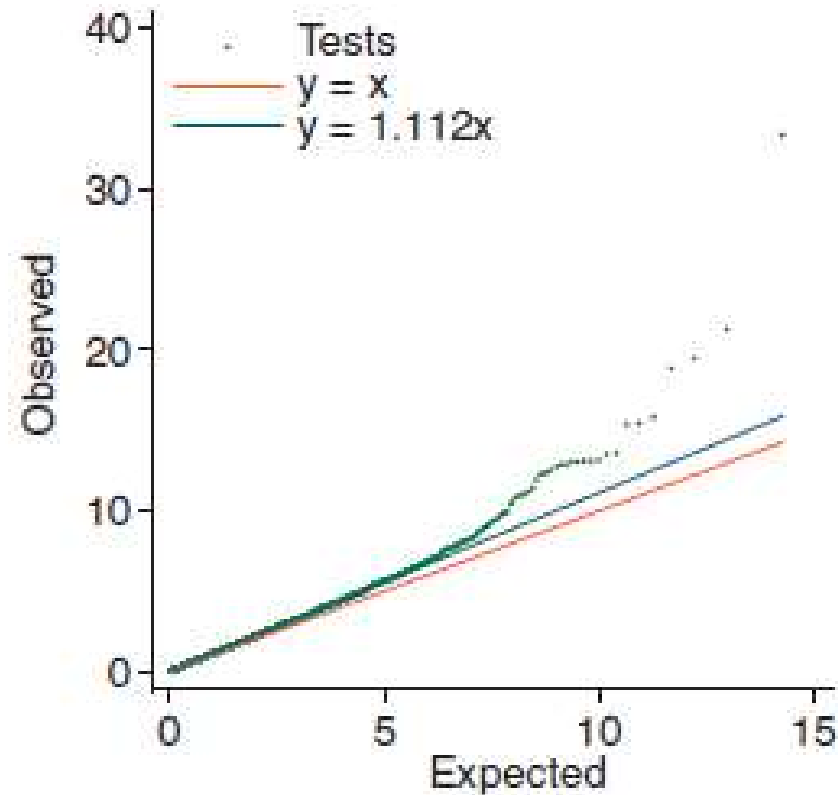
- Example: association of height in European Americans.

The height trait is associated with the lactase SNP



Spurious association due to stratification!!

2. How to **measure** population stratification



- Genomic control $\lambda = \frac{\text{median observed } \chi^2}{\text{median expected } \chi^2}$

3. How to **correct** for population stratification?

A. Genomic control

Genomic control $\lambda = \frac{\text{median observed } \chi^2}{\text{median expected } \chi^2}$

Corrected chisq statistic = chisq statistic / λ

This ensures $\lambda_{\text{GC corrected}} = 1$

Problem: insufficient correction in highly differentiated SNPs, unnecessary correction in other SNPs.

3. How to **correct** for population stratification?

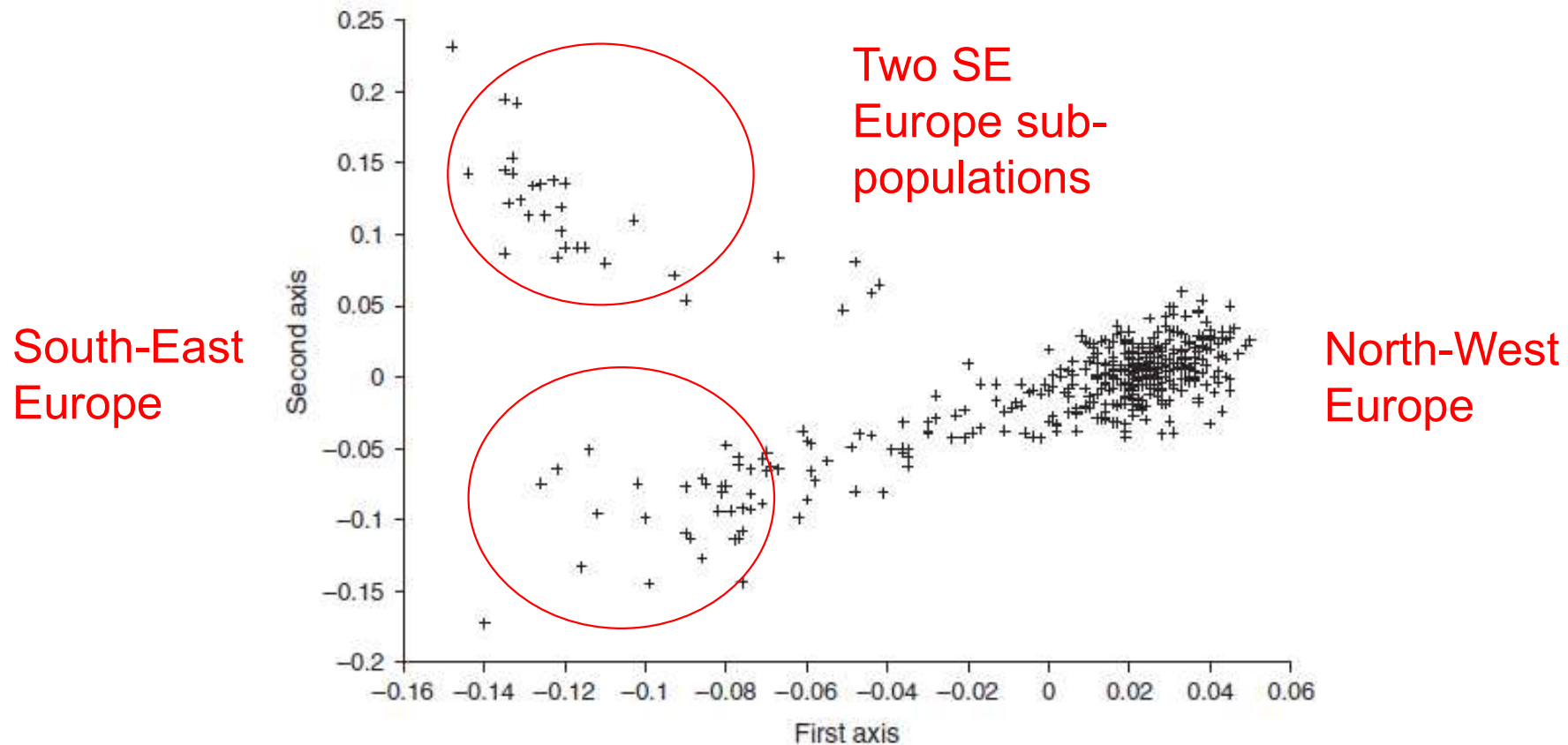
B. Inferring genetic ancestry

- STRAT: Pritchard et al. Am. J. Hum. Genet. 2000. Do not scale very well to large datasets, thus not widely used anymore.
- EIGENSTRAT: Price et al. Nature genetics 2006. widely used.
- PLINK. Purcell et al. Am. J. Hum. Genet. 2007. widely used.

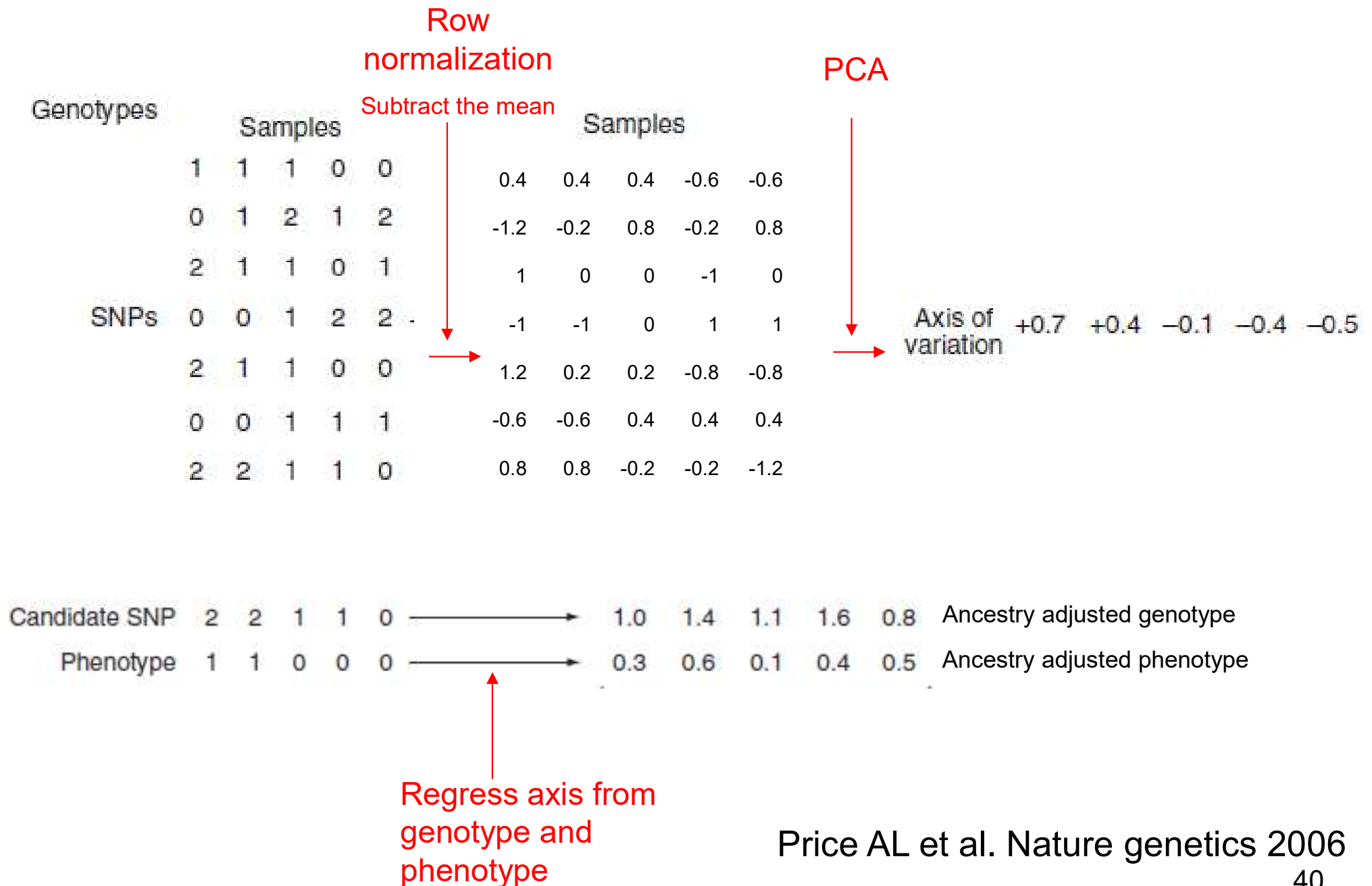
Example

		Row normalization					PCA				
Genotypes	SNPs	Samples					Samples				
		Subtract the mean									
	1	1	1	0	0	0.4	0.4	0.4	-0.6	-0.6	
	0	1	2	1	2	-1.2	-0.2	0.8	-0.2	0.8	
	2	1	1	0	1	1	0	0	-1	0	
	0	0	1	2	2	-1	-1	0	1	1	
	2	1	1	0	0	1.2	0.2	0.2	-0.8	-0.8	
	0	0	1	1	1	-0.6	-0.6	0.4	0.4	0.4	
	2	2	1	1	0	0.8	0.8	-0.2	-0.2	-1.2	
											Axis of variation
											+0.7 +0.4 -0.1 -0.4 -0.5

Top two axes of variation of European American samples



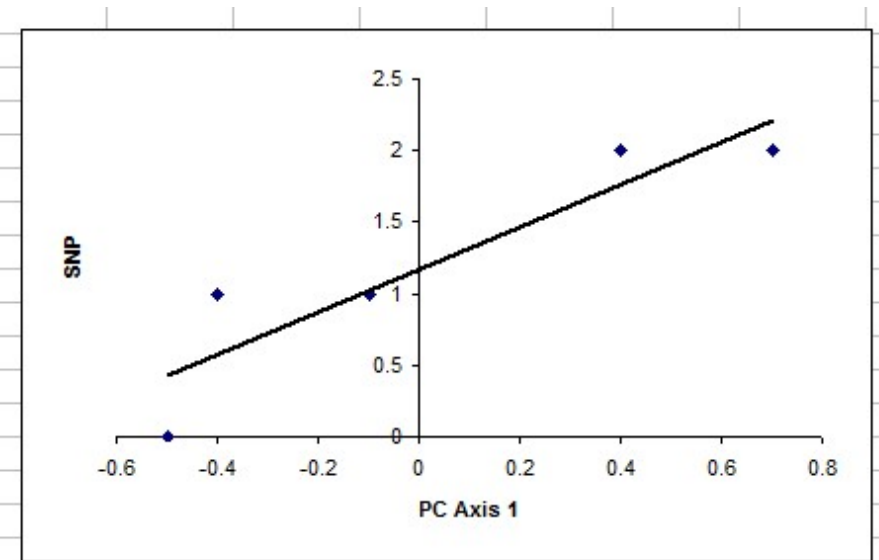
The EIGENSTRAT algorithm



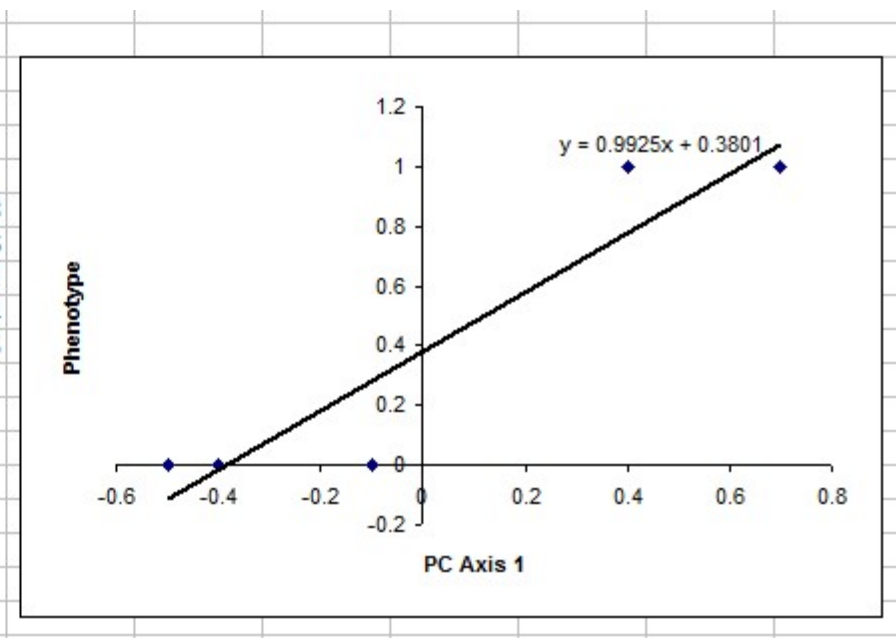
Price AL et al. Nature genetics 2006

Regress axis from genotype and phenotype

	Axis 1	candidate SNP	residuals	adjust average	
individual 1	0.7	2	-0.205992509	0.994007491	1
individual 2	0.4	2	0.237827715	1.437827715	1.4
individual 3	-0.1	1	-0.02247191	1.17752809	1.1
individual 4	-0.4	1	0.421348315	1.621348315	1.6
individual 5	-0.5	0	-0.43071161	0.76928839	0.8
		average = 1.2	average = 0	average = 1.2	



	Axis 1	Phenotype	residuals	adjust average	
individual 1	0.7	1	-0.074906367	0.325093633	0.3
individual 2	0.4	1	0.222846442	0.622846442	0.6
individual 3	-0.1	0	-0.280898876	0.119101124	0.1
individual 4	-0.4	0	0.016853933	0.416853933	0.4
individual 5	-0.5	0	0.116104869	0.516104869	0.5
		average = 0.4	average = 0	average = 0.4	



How many SNPs are required?

A small number of SNPs (a few hundreds) are not sufficient to correct for stratification

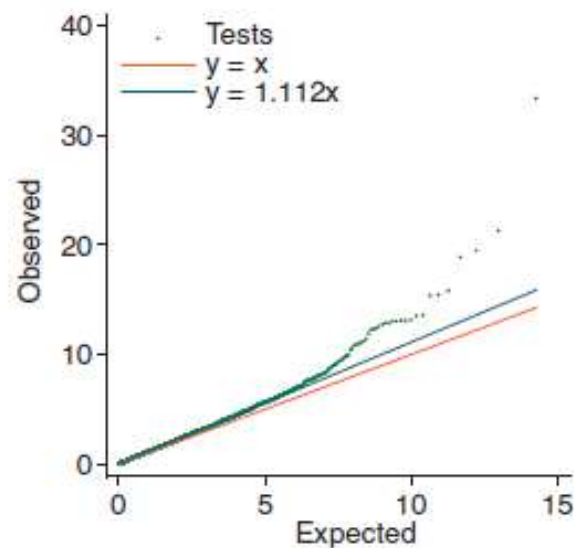
- Solution 1: correcting in GWAS with thousands of SNPs
- Solution 2: genotype a few hundreds of ancestry-informative markers (Hoggart et al. AJHG 2003)

Major questions in GWAS

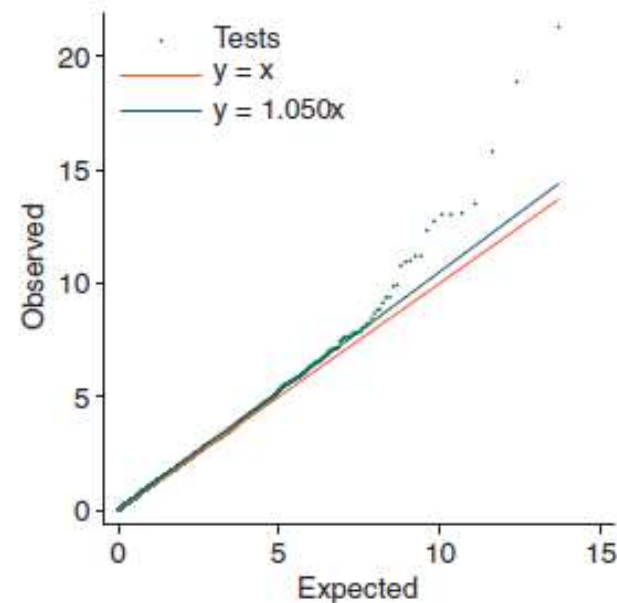
- Does Q-Q plot looks OK? (no inflation?)
- Does λ_{GC} get closer to 1 under quality filters?
- Does λ_{GC} get closer to 1 under correction for stratification?
- Do we have any genome-wide significant hits? ($<10^{-8}$)

Genomic control - example

- Genomic control $\lambda = \frac{\text{median observed } \chi^2}{\text{median expected } \chi^2}$
- Gives a measure of whether the entire distribution is inflated $\lambda > 1$ or inflated $\lambda < 1$
- In most GWAS, 1.05 is good.

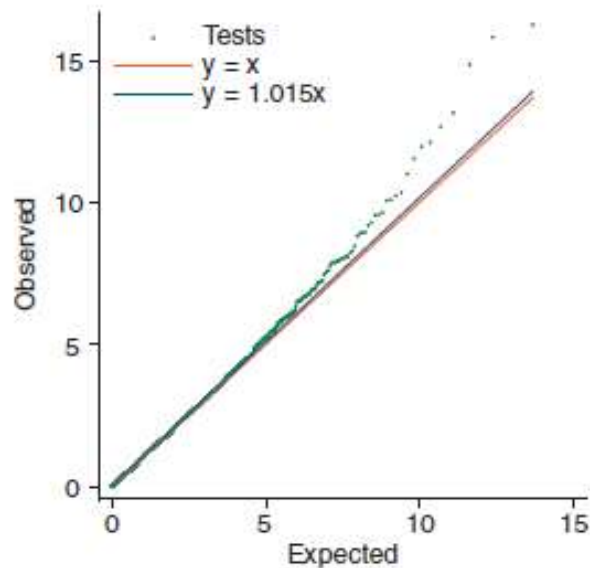


All 6322 genes
 $\lambda = 1.112$



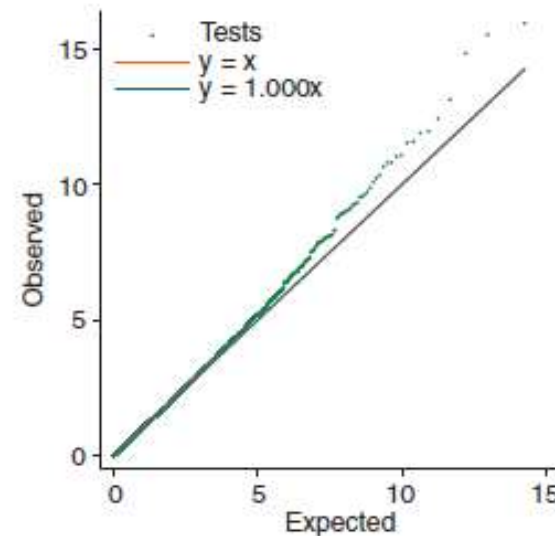
4629 genes with low difference in
call rate between case and control.
 $\lambda = 1.05$

Genomic control - example



All SNPs after stratification.

$$\lambda = 1.015$$



Filter by call rate + stratification.

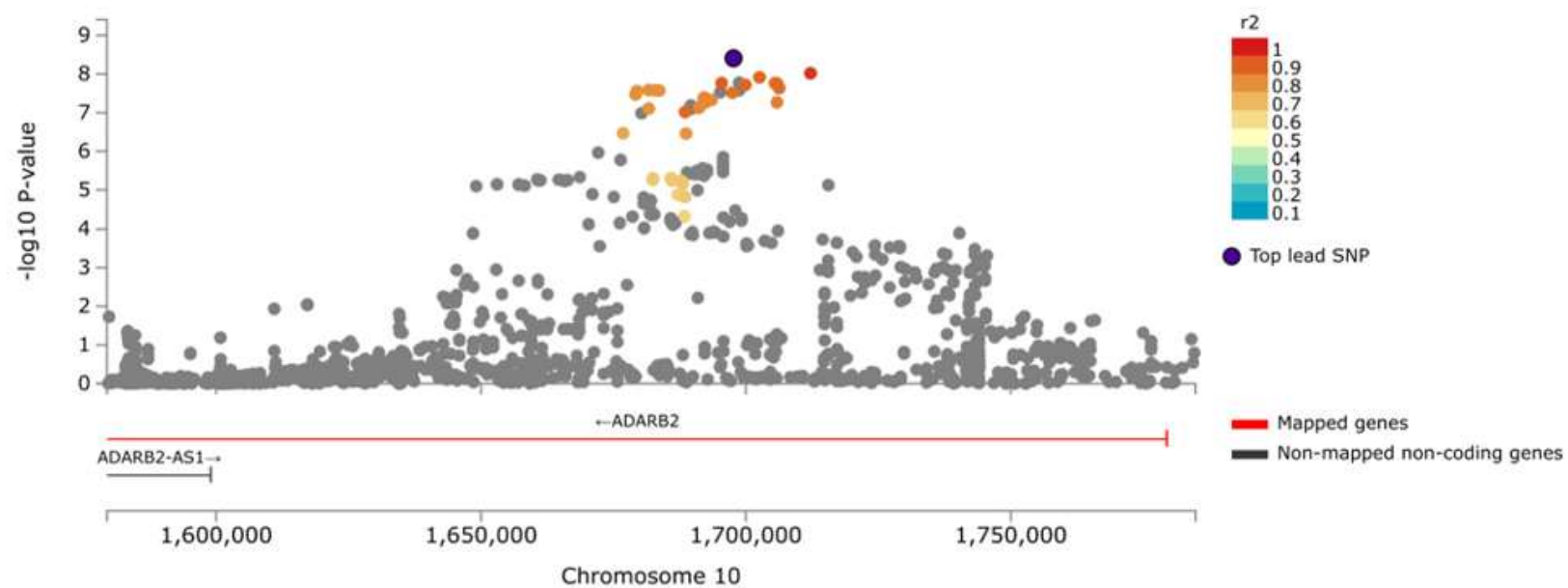
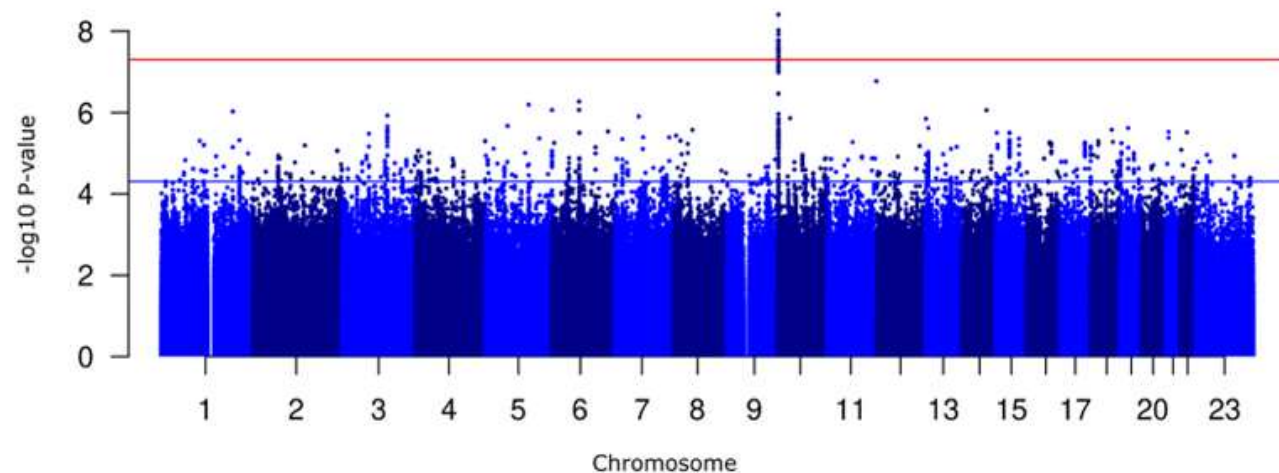
$$\lambda = 1.000$$

<https://doi.org/10.1038/s42003-024-06299-y>

Genome-wide association study reveals a locus in *ADARB2* for complete freedom from headache in Danish Blood Donors

Headache disorders are the most common disorders of the nervous system. The lifetime prevalence of headache disorders show that some individuals never experience headache. The etiology of complete freedom from headache is not known. To assess genetic variants associated with complete freedom from headache, we performed a genome-wide association study of individuals who have never experienced a headache. We included 63,992 individuals (2,998 individuals with complete freedom from headache and 60,994 controls) from the Danish Blood Donor Study Genomic Cohort.

Participants were included in two rounds, from 2015 to 2018 and in 2020. We discovered a genome-wide significant association, with the lead variant rs7904615[G] in *ADARB2* (EAF = 27%, OR = 1.20 [1.13–1.27], $p = 3.92 \times 10^{-9}$). The genomic locus was replicated in a non-overlapping cohort of 13,032 individuals (539 individuals with complete freedom from headache and 12,493 controls) from the Danish Blood Donor Study Genomic Cohort ($p < 0.05$, two-sided). Participants for the replication were included from 2015 to 2020. In conclusion, we show that complete freedom from headache has a genetic component, and we suggest that *ADARB2* is involved in complete freedom from headache. The genomic locus was specific for complete freedom from headache and was not associated with any primary headache disorders.



A PC-based correction that is used in GWAS:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{X}_s\beta_s + \sum_{i=1}^L \mathbf{A}_i\lambda_i v_i + \boldsymbol{\epsilon},$$

- If there are N samples, then $\mathbf{y}$ is a N -by-1 vector. In this paper, $\mathbf{y}$ is a binary phenotype corresponding to case/control status (see main paper for discussion of application of a linear model to binary phenotypes).
- $\mathbf{X}$ is a matrix of known covariates, such as age, sex, batch, etc.
- $\mathbf{X}_s$ is a N -by-1 vector of the genetics values at site s across all samples. Site s is being tested for its univariate association with $\mathbf{y}$.
- $\mathbf{G}$ is a N -by- M matrix whose (i, j) entry is the genetics value of marker j in sample i . In practice, we assume that the markers are normalized so that the columns of $\mathbf{G}$ have zero mean and unit variance.

By applying PCA on the $\mathbf{G}$ (the genetics matrix), $\mathbf{A}_i$ is the matrix of eigenvalues and λ_i is the eigenvalues. The Eigenvectors represent the new set of axes of the Principal component space, and the Eigenvalues carry the information of the amount of variance that each eigenvector has.

2 Linear mixed models

LMMs

In a linear mixed model (LMM), the phenotype $\mathbf{y}$ is expressed as a sum of fixed effects $\boldsymbol{\beta}, \beta_s$ and random effects $\mathbf{u}$

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{X}_s\beta_s + \frac{1}{\sqrt{M}}\mathbf{G}\mathbf{u} + \boldsymbol{\epsilon}. \quad (1)$$

- If there are N samples, then $\mathbf{y}$ is a N -by-1 vector. In this paper, $\mathbf{y}$ is a binary phenotype corresponding to case/control status (see main paper for discussion of application of a linear model to binary phenotypes).
- $\mathbf{X}$ is a matrix of known covariates, such as age, sex, batch, etc.
- $\mathbf{X}_s$ is a N -by-1 vector of the `genetics` values at site s across all samples. Site s is being tested for its univariate association with $\mathbf{y}$.
- $\mathbf{G}$ is a N -by- M matrix whose (i, j) entry is the `genetic` value of marker j in sample i . In practice, we assume that the markers are normalized so that the columns of $\mathbf{G}$ have zero mean and unit variance.
- $\mathbf{u}$ is a M -by-1 vector whose entries are identically and independently distributed (iid) samples from a Gaussian: $\mathbf{u} \sim N(0; \sigma_g^2 \mathbf{I}_M)$, where $\mathbf{I}_M$ is the M -by- M identity matrix.
- $\boldsymbol{\epsilon} \sim N(0; \sigma_e^2 \mathbf{I}_N)$ is iid random noise.

The likelihood ratio test using the full LMMs

We would like to estimate β_s for every genetic site in the data, and compute the P value for its significance. The null model is that $\beta_s = 0$ and

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \frac{1}{\sqrt{M}}\mathbf{G}\mathbf{u} + \boldsymbol{\epsilon}. \quad (2)$$

The p value is computed by comparing the ratio of the alternative and the null likelihoods to the χ^2_1 distribution.

The PC-based representation is an approximation to the full LMM representation, if a low number of PCs represent a large fraction of the variance

6 The FaST-LMM-EWASher Model

FaST-LMM-EWASher augments the linear mixed model with top PCs as covariates so that it can correct for confounding even when it is strong. The model is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{X}_s\boldsymbol{\beta}_s + \sum_{i=1}^L \mathbf{A}_i\lambda_i v_i + \frac{1}{\sqrt{M}}\mathbf{G}\mathbf{u} + \boldsymbol{\epsilon} \quad (9)$$

where $\boldsymbol{\beta}$, $\boldsymbol{\beta}_s$ and $\{v_i\}_{i=1}^L$ are fixed effects and $\mathbf{u}$ is the random effect. L is the number of top PCs used. We use an iterative procedure to find the minimum L that corrects for test statistic inflation as captured by the genomic control factor λ . This corresponds to finding the simplest model, i.e.

